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Designing a regenerative pomegranate supply chain network under uncertainty: A Lagrangian relaxation approach

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ABSTRACT

Regeneration is a fundamental concept at the heart of the circular economy. This concept emphasizes increasing positive environmental impacts in addition to achieving zero waste. Inspired by this concept, in this research, a mixed-integer linear programming model is developed to design a regenerative supply chain (SC) network in the pomegranate industry under uncertainty. In the investigated network, all pomegranate waste is transported to composting centers. Then, the compost produced by the composting centers is made available to farmers, thus integrating the concept of regeneration with SC operations. A stochastic scenario-based approach is used to deal with the uncertainty of the demand parameter. In addition, a Lagrangian relaxation algorithm is applied in order to solve the problem in large sizes. The validity of the developed algorithm is examined by comparing its results with GAMS results on nine small-sized instances. The results show that the proposed algorithm provides solutions close to GAMS solutions while being faster. Finally, the applicability of the presented model is confirmed through its implementation in a pomegranate industry in Iran.

1. Introduction

Large-scale mass production, distribution, and consumption systems are the dominant paradigm in today's industries (Pascucci and Duncan, 2017). This "take-make-dispose" system is called linear economy and depends on easy access to natural resources and energy. Such production systems take resources as input and use them to produce the final product and not rarely waste and pollution (Schulte et al., 2022). Linear systems ultimately face the challenge of increased resource starvation. (Pascucci, 2020). Due to environmental issues produced by linear systems, the idea of a green supply chain (SC) developed. Green SCs aim to minimize ecological concerns, such as reducing CO2 emissions. Green SC management regards reducing global sustainability risks like unintended negative consequences on the environment in designing the SC (Khan et al., 2023; Genovese et al., 2017).

Omitting waste in a world where the yearly generation of solid wastes reaches more than two billion tons is very important (Govindan et al., 2022). In contrast to a linear resource–products–pollution industrial system, waste and pollution are not seen as unavoidable characteristics of the SCs in the circular systems. A circular system aims to keep material in circulation. It considers reusing any undesirable output of the SC. The used or broken products will be restored, remanufactured, and recycled. In the case of biodegradable products, they may also be used as compost. Circular economy drives the concept of environmental sustainability by focusing on the vision of development of self-sustaining production systems that utilize materials over and over (Genovese et al., 2017).

The sustainable economy is the result of utilizing circular rather than linear supply chains. The core idea of circular economy is creation of self-sustaining production systems in which materials are used over and over again (Genovese et al., 2017). There are two cycles in the theory of circular economy: biological and technical cycles. Optimizing biological cycles includes regenerating ecosystems by decreasing excessive extraction of natural resources. Technical cycles prolong the life of

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products by turning waste into useful resources (Sehnem et al., 2019). The circular SCs consist of closing loops of flows of materials, sharing resources, and optimizing resource consumption (Pascucci, 2020).

Two concepts of "restorative" and "regenerative" economy are at the heart of the circular systems. Restoration is "to make something well again," and regeneration is "to make it better" than an initial condition (Morseletto, 2020). The concept of regeneration can be defined as the transformation of an undesirable material (i.e. waste) into a useable material (Park et al., 2008). For example, plastic bottles can be recycled mechanically or chemically. According to Brown et al. (2018) restorative means "restoring social and ecological systems to a healthy state" and regenerative means "enabling social and ecological systems to maintain a healthy state and to evolve". As can be seen, regenerative systems are the systems that restore what has failed and improve its current state (Gordon et al., 2022). As can be seen, adding value to evolve a state as such as possible is the aim of a regenerative SC (RSC). A very new definition is that regenerative systems has been conceptualized as the "new sustainable" (Gibbons, 2020). Morseletto (2020) proposed a definition of regeneration that is very helpful in agriculture SCs: "promotion of self-renewal capacity of natural systems with the aim of reactivating ecological processes damaged or over-exploited by human action". Jayasinghe et al. (2023) provided a rich review of definitions of regenerative food systems. Fig. 1 depicts the regenerative concept in the transition from linear systems to circular systems.

Several companies are trying to implement regenerative practices (Bag and Rahman, 2024). Specially, some of them are developing the regenerative agriculture (Bag and Rahman, 2024; Röös et al., 2023). Regenerating ecosystems is a key component of the sustainable food systems (Caron et al., 2018).

Agriculture is a vital industry because it is the source of food and also is a key component in enhancing the livelihood and financial well-being of the population (Mohan et al., 2020). Regenerative agriculture constitutes many circular farming principles, that also apply to a wider set (Bless et al., 2023). A regenerative practice in food SC is to improve the ecological conditions of a farm while also producing food (Day and Cramer, 2022). From this point of view, regenerative agriculture delineates how waste can be managed to evolve a resource within a circular food system (Mpanga et al., 2021). Such practices improve the environmental as well as the social and economic aspects of food SCs (Schreefel et al., 2022).

Some researchers stated that limited financial returns for regenerative practices within certain SCs are a challenge to implementing regenerative food SCs (Dipu et al., 2022). In contrast, some authors claimed that by implementing regenerative systems, the costs of production will be decreased (Wilson et al., 2024). This makes sense specially by increasing cost of agrochemicals in previous years (Gosnell, 2022). Farmers are interested in enhancing both financial return and soil healthiness, two coinciding incentives for agriculture, food security, and the climate (Schattman et al., 2023).

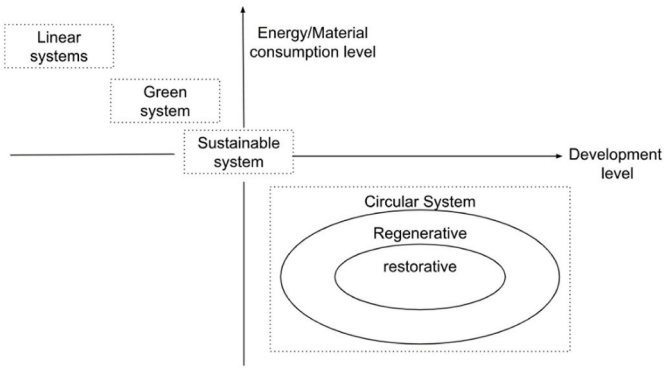


Fig. 1. Towards circular systems.

In a regenerative food system, conserving soil health state the initial step of regenerating and contributing to environmental services (Schreefel et al., 2020). These systems prioritize conserving and restoring the soil by embracing certain practices for ecosystems (Nwaogu et al., 2024). Soil conservation practices include minimal tillage, cover cropping, and compost application (Aguilar and Paulino, 2025; Zeunert, 2018). The most common soil amendment practice is composting. Composting is a common practice that has the capacity for meaningful application in the context of regenerative agriculture (Blanchard et al., 2023). The point is that there should be embedded features in the supply chain that collects and transforms the waste and other unwanted byproduct of SC to the useable products that can be used to conserver the soil health.

However, the paradigm of a regenerative food system enhances that regenerating land by adopting regenerative producers (Gordon et al., 2023). For effectible implementing the principles of regenerative food system, there needs to be a transformation towards sustainable and regenerative practices across all SC bodies and business models (Loza Adaui, 2025). It is necessary to comprehend the connections between product design, SCs, and business models as well as the stakeholders' role within the regeneration and restoration ecosystem (Amir et al., 2024).

Developing a regenerative food SC requires long-term planning (Duncan et al., 2020). Expanding food regenerative systems requires an integrated systematic approach that is important for the creation of SC (Alfina et al., 2024). It usually involves additional bodies from various sectors for implementing practices of materials recovery ecosystems (Batista et al., 2019).

For implementing circularity in a logistics network, using mathematical programming is a good device for planning (Govindan et al., 2022). Such problem formulation can integrate aspects of regenerative practices as well as strategic and operations decisions of SC design like costs associated with the operation.

With the aim of achieving zero-waste and increasing positive environmental impacts, this research introduces a RSC for agricultural products, and specifically for pomegranates. The investigated RSC is based on a closed-loop SC (CLSC) that converts pomegranate waste into compost and delivers the compost to farmers, creating positive environmental impacts. It should be noted that a mixed-integer linear programming (MILP) model under uncertainty is formulated to structure the pomegranate regenerative network, and a Lagrangian relaxation algorithm is developed to solve it. Finally, the data of a pomegranate industry in Iran will be used to validate the proposed model and algorithm.

The rest of the article is organized as follows. Section 2 deals with the literature review. The mathematical model is presented in Section 3. Section 4 is devoted to the proposed algorithm. Case study and sensitivity analysis are presented in Sections 5 and 6, respectively. Section 7 provides managerial implications. Finally, conclusion is presented in Section 8.

2. Literature review

As mentioned, in this research, a MILP model will be presented for designing a RSC for pomegranate under uncertainty. In this vein, this section concentrates on the articles wherein mathematical models have been applied to design SC networks in food and agricultural industries. Above all, the articles engaging the circularity concept are reviewed in this section with a higher interest.

Allaoui et al. (2018) presented a sustainable food SC design utilizing a two-stage multi-objective program. This considers carbon footprint costs, water footprint, and job creation in calculating the total costs of SC design. Cheraghalipour and Roghanian (2022) presented a bi-level mathematical model to design an eight-echelon CLSC. They considered various bodies including suppliers, distribution centers, customers, recycling depots, biogas centers, compost production centers, and

biogas applicants. In their article, the upper-level model aims to minimize costs while the lower-level problem aims to maximize total profits. They used two metaheuristics to solve their developed problem. Reyhani Yamchi et al. (2024) presented a MILP model for the design of a sustainable CLSC network for apple. The objectives of their model were minimizing the costs and emissions of greenhouse gases, and maximizing the justice-oriented job creation and the response to customer demand. They applied metaheuristic algorithms to solve their presented problem. Roghanian and Cheraghalipour (2019) formulated a MILP model to design a sustainable CLSC network for agricultural products. The objectives of the model were to minimize the costs and emissions of greenhouse gases, and maximize response to customer demand. Their model included centers for fresh fruit concentrates, and vermicompost fertilizers. The developed model was solved using metaheuristic approaches. Cheraghalipour et al. (2018) suggested a CLSC design for the citrus industry in which the objective was to maximize responsiveness to customers' demand. They proposed a multi-objective metaheuristic algorithm for solving the problem. Goodarzian et al. (2023) presented a multi-objective model for a citrus SC design problem by incorporating production planning, inventory control, recycling, and locational decisions. They solved the model by some metaheuristic algorithms for medium- and large-sized problems. Liao et al. (2020) developed a MILP model to design six-echelon CLSC network for citrus including financial and environmental issues and minimizing the total cost and air pollution. For controlling and reducing environmental pollution, taxes on emissions were incorporated into the model. Metaheuristic algorithms were employed to solve the proposed model in their article. Gholian-Jouybari et al. (2023a) presented a circular saffron SC that reduces carbon emissions by utilizing agri-food waste management. They formulated it via a stochastic multi-objective programming model. Then, they applied a convex robust optimization approach that handles uncertain parameters like production capacity and saffron demand. Gholian-Jouybari et al. (2023b) formulated a MILP model for the design of a circular CLSC for the soybean industry, and solved it via some hybrid metaheuristic algorithms. Similarly, Gholian-Jouybari et al. (2024) proposed a MILP model to form a sustainable coconut SC network. They took into account both forward and reverse movements. Their model optimized the net present value of the total cost while also optimizing pollution costs and job opportunities. Gilani and Sahebi (2021) proposed a model for designing a green pistachio supply network considering pistachio's by-products, and carbon emissions. They employed a robust possibilistic programming approach to handle the uncertainties. Mirzaei et al. (2024) developed a CLSC network for managing the pistachio SC under uncertainty. The authors formulated a MILP model that considered transportation and production costs, and carbon emissions taxes. Their model took into account compost production centers and pistachio farmers. It is considered that demand depends on variables such as freshness and price. Bolívar et al. (2025) presented a MILP model for optimizing strategic and operational decisions in the SC of perishable agricultural products. They applied a location-inventory problem to structure their network and used data from a case study in Colima to validate their model. With the aim of minimizing the total costs of fruit SC network, a mixed-integer nonlinear programming (MINLP) model was formulated by Yuvaraj et al. (2025) considering multi-compartment reefer trucks (MCRT). They used data from five fruits, including mango, grapes, apple, pomegranate, and orange in India to investigate the applicability of their model. The results showed that employing the MCRT strategy resulted in significant cost reduction. A robust optimization approach was employed to solve the model under uncertainty. Rajabi-Kafshgar et al. (2023) developed a MILP model to form a pistachios CLSC network. Some metaheuristic algorithms were used to solve their problem. Salehi-Amiri et al. (2022) introduced a bi-objective model to structure a CLSC in the avocado industry. They considered the operating costs and the social factors of employment opportunities. Their model was solved by exact methods. Seydanlou et al. (2022) proposed a multi-objective optimization model for a sustainable CLSC network design in the olive industry. The objectives of their model were to minimize the total costs and carbon emissions, and maximize job opportunities. Five metaheuristics are used to solve their problem. Chouhan et al. (2021) suggested a multi-echelon sugarcane SC network to handle its by-products. They developed some hybrid metaheuristics to solve the problem. Banasik et al. (2017) presented a multi-objective MILP model for CLSC in a mushroom industry considering economic and environmental indicators including recycling technologies. Salehi-Amiri et al. (2021) proposed a CLSC network designed for the walnut industry. The network considered both forward and reverse flows. It should be noted that their problem was solved via metaheuristic algorithms. Rahemi et al. (2020) developed a bioethanol SC network model considering the competition between food and biomass feedstocks. Their proposed model considered strategic variables like decisions about locations and capacities of facilities as well as tactical decisions like inventory and production planning. Yavari and Zaker (2019) proposed a model for the design of a resilient green CLSC network for perishable goods under risk assumptions. The model aimed to mitigate the power disruption risks. Mosallanezhad et al. (2021) investigated the configuration of a shrimp CLSC network. Their network design included shrimp waste powder factories, and shrimp waste powder markets. They solved their problem using the metaheuristic algorithms. Similarly, Fasihi et al. (2021) examined a CLSC in fish industry. Abad et al. (2023) presented a bi-objective model for the design of a green closed-loop value chain-based pomegranate fruit SC under dynamic uncertainty. Their model consists of seven processing facilities including grading and sorting center, thermometrical conservation facilities, juice plants, concentrate plants, packaging and distribution centers, oil plants, and leaf plants. In their proposed model, vermicompost replaced by biochar. They considered uncertainty in the supply side of the network. Their objective was to minimize the net present value of all costs, including those from fuel consumption and carbon emissions. It is noteworthy that they used hybrid risk-neutral and risk-averse multi-stage stochastic programming approach to cope with supply uncertainty. One of the limitations of their research was the lack of development of a heuristic or meta-heuristic algorithm for solving large-scale problems, which is suggested to future researchers. In this vein, Gholipour et al. (2023) presented a bi-objective model for the design of a green CLSC under dynamic uncertainty in pomegranate industry. Their study distinguished itself from other studies by considering the costs of using artificial intelligence in the production and in the reverse logistics sector, converting pomegranate waste into recycled products including ethanol for car fuel and organic fertilizer production. It is worth noting that they have not taken into account demand uncertainty. They applied some conventional meta-heuristic models and also proposed a new hybrid meta-heuristic to solve their multi-objective model. Similarly, Gholipour et al. (2024) proposed a design for a CLSC for pomegranates. Their SC takes into account recycling centers (compost centers) and compost markets. They assumed that the wasted pomegranates would also be recycled into by-products like automotive fuel, renewable energy, and compost.

In this article, for the first time, a RSC in the pomegranate industry is designed by applying the regeneration concept. Also, a Lagrangian relaxation algorithm is developed to solve the model in large sizes. It should be noted that the efficiency of the proposed model and algorithm will be evaluated using data from a real pomegranate SC in Iran. Table 1 provides a summary of the reviewed articles to illustrate research gap. As seen in Table 1, some articles have moved towards the design of an RSC by including composting center in their network. Some of these articles have taken a step further and designed a fully RSC by turning all their waste into compost. On the other hand, although some articles have considered demand uncertainty in their SC network structuring, none of them have used the stochastic scenario-based approach to deal with demand uncertainty. In addition, some articles have not solved large-size problems, and some have used meta-heuristic algorithms to address this challenge. In other words, the use of exact-based heuristics

Table 1
A review of SC network design articles in the food and agricultural industries.

Author(s)	Model type	Solution approach	Industry	Multi-period	Multi-product	Location problem	Supplier selection	Order allocation	Composting center	Fully RSC	Stochastic demand	Large-scale problems	Exact-based heuristic	Meta-heuristic
Banasik et al. (2017)	MILP	ε-constraint method	Mushroom	*	–	*	–	–	*	–	–	–	–	–
Allaoui et al. (2018)	MILP	Pareto-based approach	General agro-food	*	*	*	*	*	–	–	–	–	–	–
Cheraghalipour et al. (2018)	MILP	Multi-objective Keshtel algorithm	Citrus	*	–	*	–	*	*	*	–	*	–	*
Roghanian and Cheraghalipour (2019)	MILP	ε-constraint method and several meta-heuristics	Citrus	*	–	*	–	*	*	*	–	*	–	*
Yavari and Zaker (2019)	MILP	LP-metric method	Dairy products	*	*	*	–	–	–	–	–	–	–	–
Rahemi et al. (2020)	MILP	Augmented ε-constraint method	Agricultural products	*	*	*	*	*	–	–	–	–	–	–
Gilani and Sahebi (2021)	MILP	Augmented ε-constraint method	Pistachio	*	*	*	*	*	–	–	–	–	–	–
Chouhan et al. (2021)	MILP	Several meta-heuristics	Sugarcane	*	*	*	*	*	*	–	–	*	–	*
Mosallanezhad et al. (2021)	MILP	Several meta-heuristics	Shrimp	–	–	*	*	*	–	–	–	*	–	*
Salehi-Amiri et al. (2021)	MILP	Several meta-heuristics	Walnut	*	*	*	–	*	–	–	–	*	–	*
Fasihi et al. (2021)	MILP	LP metric and ε-constraint methods	Fish	–	*	*	*	*	–	–	–	–	–	–
Cheraghalipour and Roghanian (2022)	MILP	Several meta-heuristics	Agricultural products	*	*	*	*	*	*	–	–	*	–	*
Salehi-Amiri et al. (2022)	MILP	LP-metric	Avocado	*	*	*	*	*	–	–	–	–	–	–
Seydanlou et al. (2022)	MILP	Several meta-heuristics	Olive	*	*	*	–	*	*	*	–	*	–	*
Abad et al. (2023)	MILP	Weighted LP-metric method	Pomegranate	*	*	*	*	*	–	–	–	–	–	–
Gholipour et al. (2023)	MILP		Pomegranate	*	*	*	–	*	*	*	–	*	–	*
Gholian-Jouybari et al. (2023a)	MILP	Modified Keshtel Algorithm	Saffron	*	–	*	*	*	–	–	–	*	–	*
Gholian-Jouybari et al. (2023b)	MILP	Several meta-heuristics	Soybean	*	*	*	*	*	*	–	–	*	–	*
Goodarzian et al. (2023)	MILP	Several meta-heuristics	Citrus	*	–	*	–	*	*	*	–	*	–	*
Rajabi-Kafshgar et al. (2023)	MILP	Several meta-heuristics	Pistachio	–	*	*	–	*	*	*	–	*	–	*
Gholian-Jouybari et al. (2024)	MILP	Non-Dominated Sorting Genetic Algorithm-II (NSGA-II)	Coconut	*	*	*	–	*	*	–	–	*	–	*
Gholipour et al. (2024)	MILP	Several meta-heuristics	Pomegranate	*	*	*	–	*	*	–	–	*	–	*
Mirzaei et al. (2024)	MILP	Several meta-heuristics	Pistachio	*	*	*	–	*	*	–	–	*	–	*
Reyhani Yamchi et al. (2024)	MILP	Several meta-heuristics	Apple	*	*	*	–	*	*	*	–	*	–	*
Bolívar et al. (2025)	MILP	–	Agricultural products	*	*	*	*	*	–	–	–	–	–	–
Yuvaraj et al. (2025)	MINLP	–	Agricultural products	*	*	*	–	*	–	–	–	–	–	–
This article	MILP	Lagrangian relaxation algorithm	Pomegranate	*	*	*	*	*	*	*	*	*	*	–

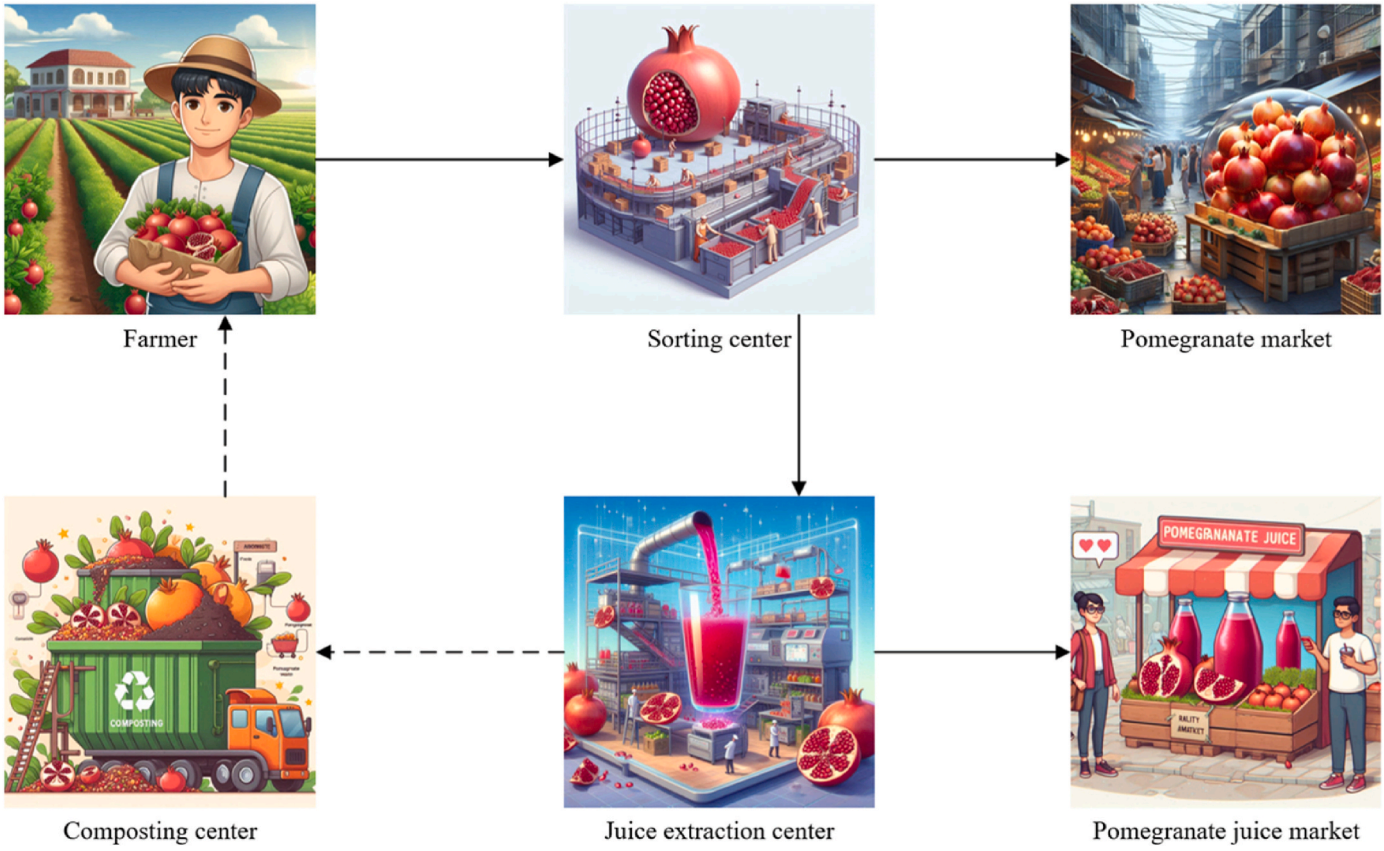


Fig. 2. The general structure of the studied RSC network.

to solve large-size problems has not been considered in this area. Hence, in this article, a novel MILP model is developed to design a fully RSC for pomegranate by considering stochastic scenario-based demand, and an exact-based heuristic leveraging the Lagrangian relaxation algorithm is developed to solve large-size problems.

3. Proposed model

In this section, a RSC for pomegranates under uncertainty is structured by integrating the stochastic scenario-based approach and a MILP model. Genovese et al. (2017) state that applying CE principles to a system leads to increased sustainability. In this article, by recycling pomegranate waste to produce compost, the principles of CE are considered in the design of the SC network. In this vein, a single objective MILP model is formulated in which CE principles are included in its constraints. The studied network is multi-period and multi-products, and includes echelons of farmers, sorting centers, pomegranate markets, pomegranate juice extraction centers, pomegranate juice markets, and composting centers. Pomegranates purchased from farmers are shipped to sorting centers to categorize. A portion is transferred to the pomegranate market, while the remainder goes to juice extraction centers. Pomegranate juice produced in juice extraction centers is sent to the market, and pomegranate waste is moved to composting centers. Compost produced in composting centers is delivered to farmers to form a RSC network. Fig. 2 represents the general structure of the studied RSC network. For a detailed understanding of the studied network, the assumptions of the proposed model are presented below:

- The network under study is multi-period and multi-product.
- The network under study is closed-loop and includes the regenerative concept.
- The farmers, sorting centers, juice extraction centers, composting centers are capacitated.
- The demand parameter is considered uncertain.
- The sorting, juice extraction, and composting centers should be located by model.
- The possibility of storing pomegranates in sorting centers has been provided.
- Market demand for pomegranates is considered certain.
- Pomegranate market demand must be fully met in each time period.

The proposed model is given below:

3.1. Mathematical model

Indices

q	Pomegranate type	$1 \le q \le Q$
i	Farmer	$1 \le i \le I$
j	Pomegranate sorting center	$1 \le j \le J$
k	Pomegranate juice extraction center	$1 \le k \le K$
l	Composting center	$1 \le l \le L$
t	Time period	$1 \le t \le T$
s	Scenario	$1 \le s \le S$

Parameters

α^1_{it}	The cost of ordering to farmer i in time period t
α^2_j	The cost of opening pomegranate sorting center j
α^4_k	The cost of opening pomegranate juice extraction center k
α^6_l	The cost of opening composting center l
β^1_{qit}	The cost of buying one unit of type q pomegranate from farmer i in time period t

(*continued on next page*)

(*continued*)

β_j^2	The cost of processing one unit of pomegranate at sorting center j
β_k^4	The cost of juicing one unit of pomegranate at juice extraction center k
β_l^6	The cost of processing one unit of pomegranate waste at composting center l
δ_{qit}^1	The capacity of farmer i to supply type q pomegranates in time period t
δ_j^2	The capacity of sorting center j
δ_k^4	The capacity of juice extraction center k
δ_l^6	The capacity of composting center l
δ_{qts}^5	The market demand for type q pomegranates in time period t under scenario s
μ_q^2	Fraction of type q pomegranates that are allocated to juicing
μ^4	Fraction of pomegranate waste
μ^6	The conversion rate of pomegranate waste to compost
θ_j^2	The cost of holding one unit of pomegranate at sorting center j
λ_{ijt}^{1-2}	The cost of shipping one unit of pomegranate from farmer i to sorting center j in time period t
λ_{jt}^{2-3}	The cost of shipping one unit of pomegranate from sorting center j to pomegranate market in time period t
λ_{jkt}^{2-4}	The cost of shipping one unit of pomegranate from sorting center j to juice extraction center k in time period t
λ_{kt}^{4-5}	The cost of shipping one unit of pomegranate juice from juice extraction center k to pomegranate juice market in time period t
λ_{klt}^{4-6}	The cost of shipping one unit of pomegranate waste from juice extraction center to composting center l in time period t
λ_{lt}^{6-1}	The cost of shipping one unit of compost from composting center l to farmers in time period t
ϑ_s	The occurrence probability of the scenario s
bigm	A large number

Variables

Y_{its}^1 { 1	If farmer i is selected in time period t under scenario s
0	Otherwise
Y_j^2 { 1	If sorting center j is opened
0	Otherwise
Y_k^4 { 1	If juice extraction center k is opened
0	Otherwise
Y_l^6 { 1	If composting center l is opened
0	Otherwise
X_{qijts}^{1-2}	The amount of type q pomegranates purchased from farmer i by sorting center j in time period t under scenario s
X_{qjts}^{2-3}	The amount of type q pomegranates shipped from sorting center j to pomegranate market in time period t under scenario s
X_{qjkts}^{2-4}	The amount of type q pomegranates shipped from sorting center j to juice extraction center k in time period t under scenario s
X_{kts}^{4-5}	The amount of pomegranate juice shipped from juice extraction center k to market in time period t under scenario s
X_{klts}^{4-6}	The amount of pomegranate waste shipped from juice extraction center k to composting center l in time period t under scenario s
X_{lts}^{6-1}	The amount of compost shipped from composting center l to farmers in time period t under scenario s
X_{qjts}	The amount of type q pomegranates stored at warehouse of sorting center j in time period t under scenario s

3.2. Objective functions

$$\sum_j X_{qijts}^{1-2} \le \delta_{qit}^1 \quad \forall q,i,t,s \tag{2}$$

$$\sum_{i,q} X_{qijts}^{1-2} \le \delta_j^2 \quad \forall j,t,s \tag{3}$$

$$\sum_{q,j} X_{qjkts}^{2-4} \le \delta_k^4 \quad \forall k,t,s \tag{4}$$

$$\sum_k X_{klts}^{4-6} \le \delta_l^6 \quad \forall l,t,s \tag{5}$$

$$X_{qjts} = \sum_i X_{qijts}^{1-2} - X_{qjts}^{2-3} - \sum_k X_{qjkts}^{2-4} \quad \forall q,j,t=1,s \tag{6}$$

$$X_{qjts} = X_{qj(t-1)s} + \sum_i X_{qijts}^{1-2} - X_{qjts}^{2-3} - \sum_k X_{qjkts}^{2-4} \quad \forall q,j,t>1,s \tag{7}$$

$$\sum_j X_{qjts}^{2-3} = \delta_{qts}^5 \quad \forall q,t,s \tag{8}$$

$$\sum_k X_{qjkts}^{2-4} = \sum_i \mu_q^2 \times X_{qijts}^{1-2} \quad \forall q,j,t,s \tag{9}$$

$$\sum_{q,j} X_{qjkts}^{2-4} = X_{kts}^{4-5} + \sum_l X_{klts}^{4-6} \quad \forall k,t,s \tag{10}$$

$$\sum_l X_{klts}^{4-6} = \mu^4 \times \sum_{q,j} X_{qjkts}^{2-4} \quad \forall k,t,s \tag{11}$$

$$X_{lts}^{6-1} = \mu^6 \times \sum_k X_{klts}^{4-6} \quad \forall l,t,s \tag{12}$$

$$\sum_{q,j} X_{qijts}^{1-2} \le bigm \times Y_{its}^1 \quad \forall i,t,s \tag{13}$$

$$\sum_{q,i} X_{qijts}^{1-2} + \sum_q X_{qjts}^{2-3} + \sum_{q,k} X_{qjkts}^{2-4} \le bigm \times Y_j^2 \quad \forall j,t,s \tag{14}$$

$$\sum_{q,j} X_{qjkts}^{2-4} + X_{kts}^{4-5} + \sum_l X_{klts}^{4-6} \le bigm \times Y_k^4 \quad \forall k,t,s \tag{15}$$

$$\sum_k X_{klts}^{4-6} + X_{lts}^{6-1} \le bigm \times Y_l^6 \quad \forall l,t,s \tag{16}$$

The objective function of the proposed model minimizes network costs, including the cost of ordering to the farmers, the cost of opening the sorting, juice extraction, and composting centers, the cost of buying pomegranates from the farmers, the cost of processing pomegranates at the sorting centers, the cost of juicing pomegranates at the juice extraction centers, the cost of processing pomegranate waste at the composting centers, the cost of storing pomegranates at the sorting centers, and the cost of shipping products between echelons.

$$\begin{aligned} Min\ Z = & \sum_{i,t,s} \vartheta_s \times \alpha_{it}^1 \times Y_{its}^1 + \sum_j \alpha_j^2 \times Y_j^2 + \sum_k \alpha_k^4 \times Y_k^4 + \sum_l \alpha_l^6 \times Y_l^6 + \sum_{q,i,j,t,s} \vartheta_s \times \beta_{qit}^1 \times X_{qijts}^{1-2} + \\ & \sum_{q,i,j,t,s} \vartheta_s \times \beta_j^2 \times X_{qijts}^{1-2} + \sum_{q,j,k,t,s} \vartheta_s \times \beta_k^4 \times X_{qjkts}^{2-4} + \sum_{k,l,t,s} \vartheta_s \times \beta_l^6 \times X_{klts}^{4-6} + \sum_{q,j,t,s} \vartheta_s \times \theta_j^2 \times X_{qjts} + \\ & \sum_{q,i,j,t,s} \vartheta_s \times \lambda_{ijt}^{1-2} \times X_{qijts}^{1-2} + \sum_{q,j,t,s} \vartheta_s \times \lambda_{jt}^{2-3} \times X_{qjts}^{2-3} + \sum_{q,j,k,t,s} \vartheta_s \times \lambda_{jkt}^{2-4} \times X_{qjkts}^{2-4} + \sum_{k,t,s} \vartheta_s \times \lambda_{kt}^{4-5} \times X_{kts}^{4-5} + \\ & \sum_{k,l,t,s} \vartheta_s \times \lambda_{klt}^{4-6} \times X_{klts}^{4-6} + \sum_{l,t,s} \vartheta_s \times \lambda_{lt}^{6-1} \times X_{lts}^{6-1} \end{aligned} \tag{1}$$

$$s.t.$$

Constraint (2) ensures that the amount of pomegranates purchased from each farmer does not exceed capacity of that farmer. Not exceeding the capacity of sorting, juice extraction, and composting centers are given in constraints (3) to (5), respectively. The inventory balance at the warehouses of sorting centers for the first and subsequent time periods are provided in constraints (6) and (7), respectively. Constraint (8) ensures that the market demand for pomegranates is met in all time periods. Constraint (9) calculates the amount of pomegranates shipped from the sorting centers to the juice extraction centers. Constraint (10) shows the inventory balance at the juice extraction centers. The amount of pomegranate waste shipped from the juice extraction centers to the composting centers is calculated by constraint (11). Constraint (12) determines the amount of compost shipped from the composting centers to the farmers. Pomegranates can only be shipped from a farmer to sorting centers if that farmer has been selected. This condition is satisfied by constraint (13). Constraint (14) guarantees that if a sorting center is not opened, that sorting center should not provide service. Similarly, this condition for juice extraction and composting centers is presented in constraints (15) and (16), respectively.

$$
\begin{aligned}
Min\ Z = & \sum_{i,t,s} \vartheta_s \times \alpha^1_{it} \times Y^1_{its} + \sum_{j} \alpha^2_j \times Y^2_j + \sum_{k} \alpha^4_k \times Y^4_k + \sum_{l} \alpha^6_l \times Y^6_l + \sum_{q,i,j,t,s} \vartheta_s \times \beta^1_{qit} \times X^{1-2}_{qijts} + \\
& \sum_{q,i,j,t,s} \vartheta_s \times \beta^2_j \times X^{1-2}_{qijts} + \sum_{q,j,k,t,s} \vartheta_s \times \beta^4_k \times X^{2-4}_{qjkts} + \sum_{k,l,t,s} \vartheta_s \times \beta^6_l \times X^{4-6}_{klts} + \sum_{q,j,t,s} \vartheta_s \times \theta^2_j \times X_{qjts} + \\
& \sum_{q,i,j,t,s} \vartheta_s \times \lambda^{1-2}_{ijt} \times X^{1-2}_{qijts} + \sum_{q,j,t,s} \vartheta_s \times \lambda^{2-3}_{jt} \times X^{2-3}_{qjts} + \sum_{q,j,k,t,s} \vartheta_s \times \lambda^{2-4}_{jkt} \times X^{2-4}_{qjkts} + \sum_{k,t,s} \vartheta_s \times \lambda^{4-5}_{kt} \times X^{4-5}_{kts} + \\
& \sum_{k,l,t,s} \vartheta_s \times \lambda^{4-6}_{klt} \times X^{4-6}_{klts} + \sum_{l,t,s} \vartheta_s \times \lambda^{6-1}_{lt} \times X^{6-1}_{lts} + \sum_{q,i,t,s} LC^1_{qits} \times \left(\sum_{j} X^{1-2}_{qijts} - \delta^1_{qit} \right) + \\
& \sum_{j,t,s} LC^2_{jts} \times \left(\sum_{i,q} X^{1-2}_{qijts} - \delta^2_j \right) + \sum_{k,t,s} LC^3_{kts} \times \left(\sum_{q,j} X^{2-4}_{qjkts} - \delta^4_k \right) + \sum_{l,t,s} LC^4_{lts} \times \left(\sum_{k} X^{4-6}_{klts} - \delta^6_l \right) \\
& s.t. \\
& \text{Constraints (6) to (16)}
\end{aligned}
\tag{17}
$$

4. Lagrangian relaxation algorithm

As a strong heuristic algorithm, Lagrangian relaxation has the capability of solving large-scale MILP problems. This algorithm initially embarks on relaxing the hard constraints and, then, moves them to the objective function; thus, the computation task gets facilitated and the problems in large scales can be solved easily. The main constituent of this algorithm is iteration and aims at finding the optimal solution by removing the distance between the lower and upper bounds (Govindan et al., 2024). Accordingly, this algorithm is used in the current research to solve the proposed MILP model. Its process is as follows:

Step 1: Extracting hard constraints

This step deals with the identification of hard constraints, which are the ones that lead to an increase in the computational complexity of the proposed model. To find the hard constraints, we remove one of the model constraints and record the runtime of the relaxed model. We implement this for all constraints. Then, we identify the constraints that have the most impact on the runtime (Heidari-Fathian and Pasandideh, 2018). By performing this operation, we found that capacity constraints lead to an increase in the model complexity. By relaxing constraints (2) to (5), the runtime was significantly reduced. For example, in instance problem 5 (IP5), the runtime decreased from 395.279 s to 96.791 s by relaxing capacity constraints. Therefore, in the proposed model, constraints (2) to (5) were identified as hard constraints and must be relaxed.

Step 2: Determining a lower bound

This step deals with relaxing constraints (2) to (5) and adding them to the objective function with Lagrangian coefficients. The obtained Lagrangian dual model is shown below:

where LC^1_{qits}, LC^2_{jts}, LC^3_{kts}, and LC^4_{lts} are the Lagrangian coefficients related to constraints (2) to (5), respectively. It should be noted that the optimal value of the Lagrangian dual model is a lower bound for the original model.

Step 3: Determining an upper bound

When the solutions calculated by the Lagrangian dual model are placed in the original model, the upper bound can be computed. However, the solution extracted from the Lagrangian dual model may be infeasible for original model; therefore, the feasibility process is included in the algorithm in order to overcome this infeasibility. When constraints (2) to (5) are relaxed in the proposed model, an infeasible solution space may arise for the original model by the optimal values already obtained for variables X^{1-2}_{qijts}, X^{2-4}_{qjkts}, and X^{4-6}_{klts}. Since the objective of the proposed model is to minimize costs, by relaxing constraints (2) to (5), the model:

- selects exactly one farmer to purchase type q pomegranates in each time period,
- establishes exactly one sorting center,
- establishes exactly one juice extraction center,
- and establishes exactly one composting center.

In the process of feasibility, we make it possible to purchase type q pomegranates from multiple farmers per time period, and establish multiple sorting centers, juice extraction centers, and composting centers as needed.

Step 4: This step utilizes the sub-gradient proposed by Fisher et al. (2004) for updating the algorithm parameters. One of the key parameters of the algorithm is the step size, which plays a key role in converging to the optimal solution. If the value of this parameter is large, it may prevent convergence to the optimal solution. In contrast, if it is small, the algorithm slows down the and may stop

before reaching the optimal solution due to the iteration limit. The formulas for calculating step sizes are given in Eqs. (18)–(21).

$$SS^{1,n} = \frac{\varpi^n \times (UPR^* - LWR^{*n})}{\sum_{q,i,t,s}\left(\sum_j X^{1-2}_{qijts} - \delta^1_{qit}\right)^2} \tag{18}$$

$$SS^{2,n} = \frac{\varpi^n \times (UPR^* - LWR^{*n})}{\sum_{j,t,s}\left(\sum_{i,q} X^{1-2}_{qijts} - \delta^2_{j}\right)^2} \tag{19}$$

$$SS^{3,n} = \frac{\varpi^n \times (UPR^* - LWR^{*n})}{\sum_{k,t,s}\left(\sum_{q,j} X^{2-4}_{qjkts} - \delta^4_{k}\right)^2} \tag{20}$$

$$SS^{4,n} = \frac{\varpi^n \times (UPR^* - LWR^{*n})}{\sum_{l,t,s}\left(\sum_{k} X^{4-6}_{klts} - \delta^6_{l}\right)^2} \tag{21}$$

where UPR^* indicates the best upper bound and LWR^{*n} shows the best lower bound at iteration n. Parameter ϖ is mostly regarded within the range of 0–2. In this study, the first value of this parameter is 2; and its value will be halved if the lower bound value does not witness any improvement after three consecutive iterations. Similarly, Lagrangian coefficients are computed through Eqs. (22)–(25).

$$LC^{1,n+1}_{qits} = Max\left\{0, LC^{1,n}_{qits} + SS^{1,n} \times \left(\sum_j X^{1-2}_{qijts} - \delta^1_{qit}\right)\right\} \tag{22}$$

$$LC^{2,n+1}_{jts} = Max\left\{0, LC^{2,n}_{jts} + SS^{2,n} \times \left(\sum_{i,q} X^{1-2}_{qijts} - \delta^2_{j}\right)\right\} \tag{23}$$

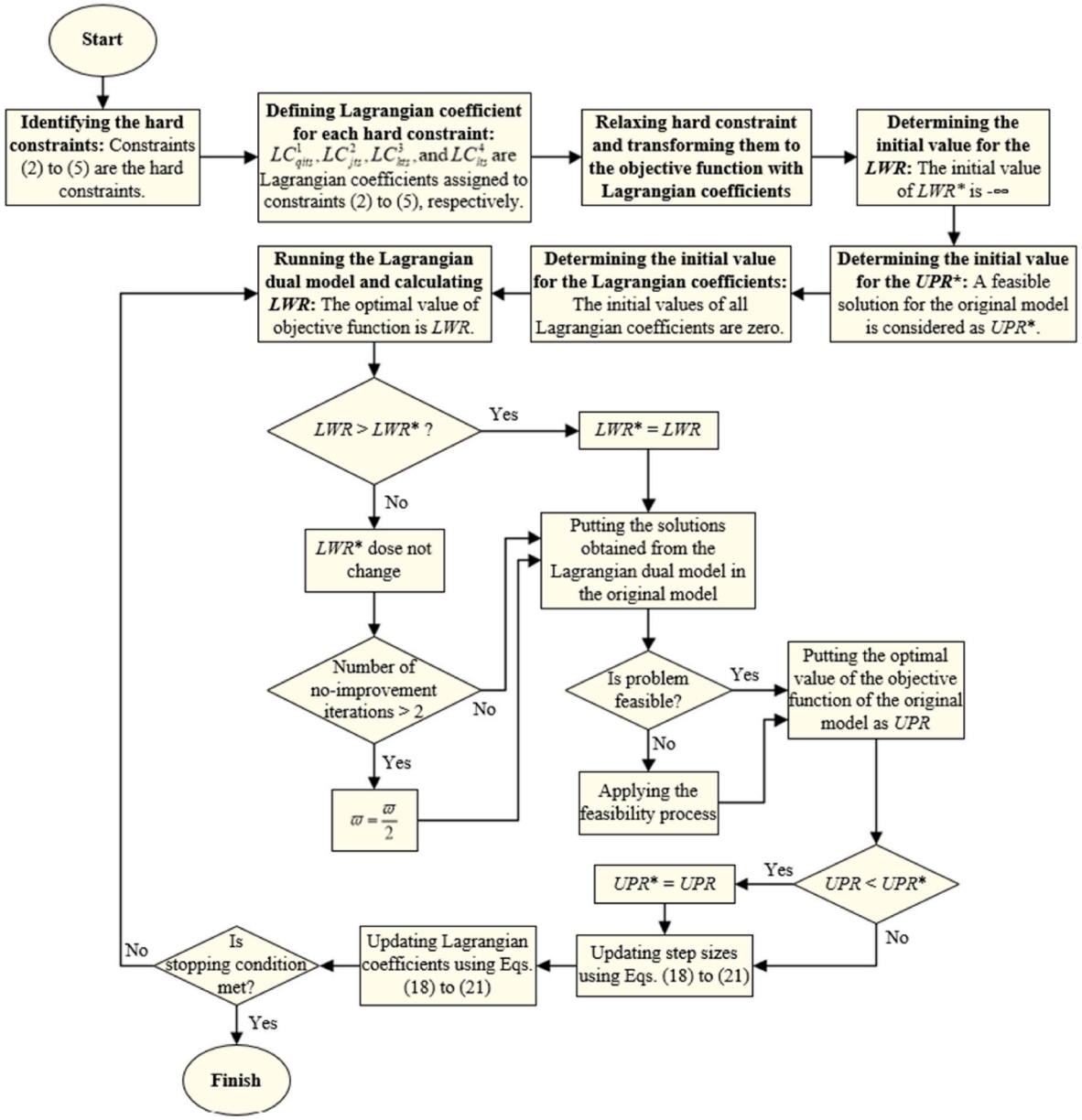


Fig. 3. The general structure of the Lagrangian relaxation algorithm used in this article.

Table 2
The data simulation functions.

Parameter	Simulation function	Parameter	Simulation function
α_{it}^{1}	$Round(uniform(500,550))$	δ_{qts}^{5}	$\begin{cases} J \times Round(uniform(7000,8500)) & q=1, s=1 \\ Round\left(1.1 \times \delta_{qt(s-1)}^{5}\right) & q=1, s>1 \\ J \times Round(uniform(4000,5300)) & q=2, s=1 \\ Round\left(1.1 \times \delta_{qt(s-1)}^{5}\right) & q=2, s>1 \\ J \times Round(uniform(2000,2500)) & q=3, s=1 \\ Round\left(1.1 \times \delta_{qt(s-1)}^{5}\right) & q=3, s>1 \end{cases}$
α_{j}^{2}	$Round(uniform(48000,52000))$	μ_{q}^{2}	$\begin{cases} Uniform(0.14,0.18) & q=1 \\ Uniform(0.12,0.15) & q=2 \\ Uniform(0.08,0.12) & q=3 \end{cases}$
α_{k}^{4}	$Round(uniform(31000,34000))$	μ^{4}	0.45
α_{l}^{6}	$Round(uniform(16000,18000))$	μ^{6}	0.4
β_{qit}^{1}	$\begin{cases} Uniform(0.25,0.28) & q=1 \\ Uniform(0.3,0.34) & q=2 \\ Uniform(0.36,0.4) & q=3 \end{cases}$	θ_{j}^{2}	$Uniform(0.015,0.02)$
β_{j}^{2}	$Uniform(0.02,0.035)$	λ_{ijt}^{1-2}	$Uniform(0.007,0.009)$
β_{k}^{4}	$Uniform(0.025,0.04)$	λ_{jt}^{2-3}	$Uniform(0.008,0.01)$
β_{l}^{6}	$Uniform(0.03,0.04)$	λ_{jkt}^{2-4}	$Uniform(0.01,0.012)$
δ_{qit}^{1}	$\begin{cases} Round(uniform(12000,16000)) & q=1 \\ Round(uniform(7000,10000)) & q=2 \\ Round(uniform(3500,5000)) & q=3 \end{cases}$	λ_{kt}^{4-5}	$Uniform(0.008,0.01)$
δ_{j}^{2}	$Round(uniform(60000,65000))$	λ_{klt}^{4-6}	$Uniform(0.005,0.007)$
δ_{k}^{4}	$Round(uniform(15000,17000))$	λ_{lt}^{6-1}	$Uniform(0.005,0.007)$
δ_{l}^{6}	$Round(uniform(5000,6000))$	ϑ_{s}	$\begin{matrix} 0.3 & s=1 \\ 0.5 & s=2 \\ 0.2 & s=3 \end{matrix}$

$$LC_{kts}^{3,n+1} = Max\left\{0, LC_{kts}^{3,n} + SS^{3,n} \times \left(\sum_{q,j} X_{qjkts}^{2-4} - \delta_{k}^{4}\right)\right\} \quad (24)$$

$$LC_{lts}^{4,n+1} = Max\left\{0, LC_{lts}^{4,n} + SS^{4,n} \times \left(\sum_{k} X_{klts}^{4-6} - \delta_{l}^{6}\right)\right\} \quad (25)$$

In the proposed algorithm, two conditions are considered for stopping the algorithm. If one of the conditions is met, the algorithm will stop. These conditions are given below:

(1) If the gap ratio between LWR^{*} and UPR^{*} (i.e., $\left(\frac{UPR^{*}-LWR^{*}}{UPR^{*}}\right)$) gets smaller than 0.03.
(2) If the algorithm is run 60 iterations.

The general structure of the Lagrangian relaxation employed for solving the proposed MILP model is shown in Fig. 3.

Table 3
The size of simulated problems.

Size	Instance problem	Indices						
		q	i	j	k	l	t	s
Small	IP1	3	6	3	2	2	6	3
	IP2	3	8	4	3	3	8	3
	IP3	3	10	5	4	4	10	3
	IP4	3	12	6	4	4	12	3
	IP5	3	13	6	5	5	14	3
	IP6	3	14	7	5	5	16	3
	IP7	3	15	7	6	6	18	3
	IP8	3	16	8	6	6	20	3
	IP9	3	17	8	7	7	22	3
Large	IP10	3	18	9	7	7	24	3
	IP11	3	19	9	8	8	26	3
	IP12	3	20	10	8	8	28	3
	IP13	3	21	10	9	9	30	3
	IP14	3	22	11	9	9	32	3

5. Case study

Saveh city with 10 thousand hectares of pomegranate orchards is one of the largest pomegranate producers in Iran and produces more than 170 thousand tons of pomegranates annually. This city is located 100 km from Tehran (the capital of Iran). In this research, we will use the data of a pomegranate SC in Saveh to validate our proposed model and algorithm. Due to the incompleteness of historical data and the lack of classification for the existing data, we applied probabilistic distribution functions and integrated them with expert knowledge to simulate real-world data. For this purpose, the knowledge of three experts, including an agricultural engineer with more than 10 years of experience in the pomegranate industry, a manager of a local agricultural cooperative company, and a manager of agricultural jihad ministry specializing in agricultural data and statistics was used. To simulate the data, an online meeting was first held and the parameters of the proposed model were explained in detail to the experts. Then, they were asked to prepare a lower bound and an upper bound for each parameter for the next session by analyzing the parameters. It is noteworthy that to increase the speed of data extraction and prevent experts from getting tired, the parameters were categorized and each category was assigned to an expert. In the second online meeting, after collecting the lower and upper bounds of the parameters from the experts, they were asked to exchange ideas on some key parameters such as δ_{qts}^{5}, μ_{q}^{2}, μ^{4}, and μ^{6} in order to achieve convergence. Finally, uniform distribution functions were formulated using extracted lower and upper bounds of parameters to simulate the data. The data simulation functions are presented in Table 2.

It should be noted that market demand, as well as the capacity of farmers and centers, is considered in kilograms, while all costs are stated in dollars. To investigate the performance of the developed Lagrangian relaxation algorithm, we first generate nine instance problems in small

Table 4
The comparison of GAMS and proposed algorithm in nine small-sized problems.

Instance problem	Objective function				Runtime (second)		
	GAMS	Lagrangian relaxation		Optimality gap (%)	GAMS	Lagrangian relaxation	Ratio $\left(\frac{\text{LR}}{\text{GAMS}}\right)$
		*LWR**	*UPR**				
IP1	268,205	267,126	268,205	0	3.852	4.587	1.19
IP2	358,758	356,862	358,758	0	12.750	10.134	0.79
IP3	505,058	500,348	509,635	0.91 %	28.233	22.136	0.78
IP4	728,681	719,004	736,912	1.13 %	84.209	52.361	0.62
IP5	768,919	757,706	779,653	1.40 %	395.279	127.014	0.32
IP6	1,005,510	962,653	1,026,287	2.07 %	943.445	305.167	0.32
IP7	1,090,147	1,049,566	1,116,738	2.44 %	1198.008	456.561	0.38
IP8	1,315,253	1,258,398	1,354,620	2.99 %	1421.793	503.284	0.35
IP9	1,424,017	1,386,413	1,476,374	3.68 %	1916.186	647.920	0.34
Average				1.62 %	667.084	236.574	0.35

Table 5
The comparison of GAMS and proposed algorithm in five large-sized problems.

Instance problem	Objective function				Runtime (second)	
	GAMS	Lagrangian relaxation		Feasibility gap (%)	GAMS	Lagrangian relaxation
		*LWR**	*UPR**			
P10	1,801,149	1,580,944	1717474	4.87 %	2000	795.003
P11	2014235	1742862	1883375	6.95 %	2000	911.270
P12	2416376	2066173	2214746	9.10 %	2000	1232.481
P13	2497116	2151280	2330884	7.13 %	2000	1404.708
P14	3013628	2598471	2732688	10.28 %	2000	1732.564
Average				7.67 %	2000	1215.205

sizes and five instance problems in large sizes by the simulation functions shown in Table 2. Then, on the small instance problems, we compare the results of the Lagrangian relaxation algorithm with the results of GAMS. In this paper, problems that GAMS solves in less than 2000 s are considered as small instance problems. Table 3 shows the size of the simulated problems. It should be noted that instance problem 13 (IP13) is the case study of this article.

With the purpose of validating the proposed algorithm, we have compared its results with those obtained from GAMS. In this vein, nine small-sized instance problems are solved once by the CEPLEX solver in GAMS software, and run again by the developed Lagrangian relaxation algorithm. Table 4 reports the results of Lagrangian relaxation and GAMS for nine small-sized instance problems.

$$\text{Optimality gap} = \frac{UPR^{*} - \text{GAMS}}{\text{GAMS}} \times 100$$

Analysis of the results presented in Table 4 indicates the effectiveness of the Lagrangian relaxation algorithm. As can be seen, the solutions of the proposed algorithm and GAMS in IP1 and IP2 are exactly the same, and the average optimality gap for the nine instance problems is 1.62 %. On the other hand, comparing the runtimes of GAMS and Lagrangian relaxation algorithm reveals that as the problem size increases, the runtime of the proposed algorithm decreases significantly compared to the runtime of GAMS. This means that the developed algorithm is able to calculate near-optimal solutions in less time and is more efficient in large-size instance problems compared to GAMS. Therefore, by applying the two indicators (i.e., optimality gap and runtime ratio), the validity of the proposed algorithm and the reliability of its results are confirmed.

To examine the performance of the proposed Lagrangian relaxation algorithm in solving large-scale problems, we employed the proposed algorithm to solve five large-sized instance problems (i.e., IP10 to IP14) and compared its results with those obtained from GAMS. The results are shown in Table 5.

$$\text{Feasibility gap} = \frac{\text{GAMS} - UPR^{*}}{UPR^{*}} \times 100$$

Comparison of the results calculated by GAMS and the developed algorithm for five large-sized problems indicates the efficiency of the proposed algorithm. In five large-size problems, GAMS failed to reach global optimal solutions in 2000 s, but the proposed algorithm was able to calculate better solutions than GAMS in less time. In other words, in solving large-scale problems, the developed algorithm is better than GAMS in both feasibility gap and runtime.

As stated, IP13 data was generated to align with a real-world case study of a pomegranate SC in Iran. In this problem, three types of pomegranates, 21 potential suppliers, 10 potential sorting centers, nine potential juice extraction centers, nine potential composting centers, and 30 time periods are considered. Noted that the time periods are daily, and the time horizon covers one month. The following are the results of solving case study (IP13) using the proposed algorithm:

- The optimal value of objective function is 2330884.
- The sorting centers 1, 5, 6, and 10 should be opened.
- The juice extraction centers 2 and 8 should be opened.
- The composting centers 2, 5, and 8 should be opened.

Table 6 shows the selected suppliers in each time period under scenario 1. For example, it can be inferred from the table that orders should be placed with suppliers 2, 4, 8, 11, 13, 17, and 21 in the first time period. The selected suppliers for scenarios 2 and 3 are given in Tables A1 and A2, respectively.

- The amount of type 1 pomegranates purchased from suppliers by sorting centers in each time period under scenario 1 is shown in Table 7. For instance, the number 13200 in the first row of this table means that if scenario 1 occurs, 13,200 kg of type 1 pomegranates should be purchased from supplier 1 and transferred to sorting center 1 in time period 17. Also, the amounts of type 2 and 3 pomegranates purchased from suppliers by sorting centers in each time period under scenario 1 are presented in Tables A3 and A4, respectively.
- The optimal values calculated for other decision variables (i.e., X^{2-3}_{qjts}, X^{2-4}_{qjkts}, X^{4-5}_{kts}, X^{4-6}_{klts}, X^{4-6}_{klts}, and X_{qjts}) under scenario 1 are denoted in the Appendix.

Table 6
The selected suppliers for each time under scenario 1.

Y^1_{its}	Time period (t)																												
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Supplier (i) 1					✓								✓				✓				✓		✓		✓		✓	✓		✓
2	✓							✓		✓	✓		✓	✓			✓		✓	✓					✓			✓		
3		✓			✓													✓					✓			✓	✓		✓	
4	✓	✓	✓				✓		✓	✓																✓				✓
5					✓	✓									✓				✓					✓		✓		✓		
6					✓	✓					✓	✓				✓		✓		✓				✓				✓	✓	✓
7				✓			✓														✓									
8	✓			✓		✓		✓	✓				✓													✓	✓			
9		✓		✓		✓		✓	✓	✓			✓	✓						✓		✓		✓			✓			
10	✓	✓								✓				✓					✓											
11	✓	✓						✓	✓			✓					✓				✓	✓			✓		✓			
12			✓			✓								✓						✓		✓		✓						
13	✓		✓	✓			✓			✓													✓					✓	✓
14		✓	✓						✓										✓									✓	
15			✓							✓	✓						✓				✓									
16				✓							✓				✓		✓													
17	✓			✓				✓							✓							✓								
18		✓			✓											✓			✓			✓				✓				
19							✓		✓							✓						✓						✓		
20					✓					✓						✓					✓									
21			✓	✓			✓				✓										✓								✓	

6. Sensitivity analysis

In this section, the performance accuracy of the proposed model is examined by implementing a sensitivity analysis process. A review of the literature reveals that the demand parameter is appropriate for the sensitivity analysis in models similar to ours (Mehrabi et al., 2025; Taherifar et al., 2024; Tavana et al., 2022). Therefore, in this article, we implement the sensitivity analysis process on the demand parameter. If the model is formulated correctly, it is expected that the optimal value of the objective function will increase with the increase of the demand rate, and with its decrease, the optimal value of the objective function will decrease. In this vein, we define nine cases by changing the demand values of IP13 and examine the correctness of the model's performance by analyzing the results of the nine defined cases. The defined cases along with their results are given in Table 8. The value of the objective function for each case is depicted in Fig. 4.

As shown in Table 8 and Fig. 4, according to our expectations, there is a direct relationship between the increase in the demand rate and the increase in the optimal value of the objective function. The results obtained from the sensitivity analysis process confirm the accuracy of the model's performance. Therefore, it can be concluded that the model is correctly formulated and the results obtained from it are reliable.

7. Managerial implications

A major contribution of this study is that it puts focus on the RSC model that is highly effective in waste reduction and improved environmental performance in the context of pomegranate industry. After changing the pomegranate waste into compost and offering it to farmers, the loop is closed by the model while preserving efficiency and sustainability. It is noteworthy that the mentioned approach can be put into practice in other agricultural parts, including figs, pistachios, citrus fruits, and the perishable goods wherein organic waste can be used for effective purposes. In fact, the adaptability of this model provides industries with this opportunity to approach circular economy.

The results obtained from this article provide valuable practical insights for decision makers and managers and can help them in making strategic and operational decisions as a decision support system. The results obtained for the case study (IP13) showed that four out of ten potential sorting centers, three out of nine potential composting centers, and two out of nine potential juice extraction centers should be opened for structuring the network under uncertainty. The establishment of the centers is based on cost and distance criteria. This means that the distances traveled by vehicles are taken into account in the decision-making. This leads to a reduction in the total distance traveled by vehicles, which results in a reduction in carbon emissions.

Flexibility and dynamism in farmer selection is another management insight that is deduced from the results. The results presented in Tables 6, A1, and A2 indicate that under different scenarios, the set of pomegranate suppliers varies in each time period. This informs the decision makers that they should not always order from a specific number of suppliers, but should provide a large list of suppliers and choose the best suppliers based on quality, cost, and capacity in different time periods and scenarios.

It should be noted that the results of demand sensitivity analysis have presented valuable managerial insight. The results of the sensitivity analysis report useful financial information to decision makers, showing how costs are affected by changes in market demand. For example, if the company uses marketing strategies to increase demand, the results of the sensitivity analysis will provide the budget needed to increase demand. This makes the decision makers aware of the company's cost changes in case of demand side changes, and take the necessary measures to provide the required funds.

On the whole, the inclusive framework presented in this study creates a trade-off between economic objectives and environmental sustainability. The integration of this RSC model with an efficient solving

Table 7
The amount of type 1 pomegranates purchased from suppliers by sorting centers in each time period under scenario 1.

X^{1-2}_{qijt}			Time period (t)																												
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
q=1	i=1	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13200	0	0	0	0	0	0	0	9817.6	0	12480	13308	0	0
		j=5	0	0	0	0	6779.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13568	0	14113	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	13871	0	0	0	0	0	0	0	0	0	0	0	3705.4	0	0	0	0	13920
	i=2	j=1	15946	0	0	0	0	0	0	15996	0	0	0	0	13037	0	0	0	0	0	15906	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13772	0	0	14564	0	0
		j=6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2378	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	15632	0	0	12529	0	0	10078	0	0	13732	0	0	0	0	0	0	0	0	0	0
	i=3	j=1	0	0	0	15367	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14643	0	0	0	0	0	0	0
		j=5	0	14234	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12383	0	13386	0	
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12443	0	0	0	0	0	0	0	0	0	0	0	0	0
	i=4	j=5	0	0	13709	0	0	0	14385	0	14658	12894	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13937	0	0	0	3689.7	
		j=10	15114	14804.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8794.3	
	i=5	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12790.4	0	0	0	0	14883	0	13740	0	0	0	0
		j=5	0	0	0	0	3249.6	0	0	0	0	0	0	6121.5	0	0	0	0	0	0	1825.6	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	12541.4	12488	0	0	0	0	0	6215.5	0	0	15205	0	0	0	0	0	0	0	0	0	0	0	0	15961	0	
	i=6	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	2518.3	0	5550	0	0	0	0	0	0	0	0	0	0	0	0	0	7124.3
		j=5	0	0	0	0	14988	0	0	0	0	0	11333.7	13992	0	0	10728.1	0	0	6631.9	0	0	0	0	0	0	0	0	0	5277.5	7820.6	0
		j=10	0	0	0	14617	0	0	0	0	0	0	809.3	0	0	0	0	0	7871.1	0	0	0	0	0	197.6	0	0	0	0	10410.5	8053.4	8174.7
	i=7	j=1	0	0	0	0	0	0	0	3340	0	0	0	0	0	0	0	13080	0	0	0	3245.3	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	15416	0	0	13140	12335	0	0	0	0	0	0	0	0	15531	0	0	0	0	0	0	0	0	0	0	0	0	0
	i=8	j=1	0	0	0	0	13248	0	0	15370	14424	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	15181	0	
		j=5	15197	0	0	4987.5	0	0	0	0	0	0	0	0	0	0	14758	0	0	0	0	0	0	0	3095.2	0	0	0	0	0	0	0
		j=10	0	0	0	10203.5	0	15266	0	0	0	0	0	0	11387.5	0	0	0	0	0	0	0	0	0	9020.8	0	0	14589	0	12498	0	0
	i=9	j=1	0	0	0	13723	0	13391	0	0	0	15827	0	0	5672	0	0	0	0	0	0	3636.5	0	6837.8	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	11372.2	0	0	0	8335	0	0	0	0	0	0	10621.5	0	0	0	0	0	5037.7	0	0	0	
		j=10	0	13241	0	0	0	0	0	13148	1642.8	0	0	0	0	12921	0	0	0	0	0	0	0	6230.2	0	0	0	0	0	10176.5	0	0
	i=10	j=1	1088.6	15568	0	0	0	0	0	0	0	0	0	0	0	13383	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	14526.9	0	15512	0	0	0	0	0	0	0	0	12907	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	15215	15056	0	0	0	0	0	0	0	0	14808	0	15792	15886	0	0	0	0	0
	i=11	j=1	2882.4	0	0	0	0	0	0	0	14312	0	0	0	13129	0	0	0	0	0	15389	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	8771.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14036	0	13267	0	14651	0	13016	0	0	0
		j=10	11857.6	5208.8	0	0	0	0	0	0	0	0	0	0	0	0	0	14613	3805.1	0	0	15780	0	0	0	0	0	0	0	0	0	0
	i=12	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14036	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	14608	0	0	15297	0	0	0	0	0	0	0	1885	0	0	0	0	0	0	0	14609	0	13997	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	12947	0	0	0	0	0	0	15899	0	0	0	0	0	0	0	0	0
	i=13	j=1	0	0	0	13685	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14859	0	0	0	0	0	0	0	1849.8	0	
		j=5	0	0	0	0	0	0	14439	0	0	0	0	0	0	0	0	0	0	0	15128	0	0	14243	13785	0	0	0	0	0	0	14472
		j=10	13260	0	12482	0	0	0	0	0	0	13381	0	0	0	0	14873	0	0	0	0	0	0	0	0	0	0	0	0	0	11129.2	0
	i=14	j=1	0	0	7511.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13168	0	0	0	0	0	14758	0	0	0	0	15463
		j=5	0	0	4861.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12004	0	0	0	0	0	0	0	0	0	0
		j=6	0	15055	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	14552	0	0	0	0	0	0	0	13979	0	0	0	0	0	0	0	0	0	14763	0	0	0
	i=15	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5351	0	0	0	3571.3	0	0	0	0	0	0	0	0	0
		j=5	0	0	10425.9	0	0	0	0	0	0	0	9702.1	0	0	0	0	0	8993	0	0	14398	9011.7	0	0	0	11268	0	0	0	0	0
		j=10	0	0	1854.1	0	0	0	14976	0	0	12041	0	0	0	0	0	0	0	0	0	0	0	0	0	0	963	15067	13795	0	0	0
	i=16	j=1	0	0	0	10049	0	15649	401.1	0	0	0	14371	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3453.7	0	0	0	0
		j=5	0	0	3156	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13348	0	11119.8	
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13867	0	0	0	0	0	9648.3	0	0	0	0
	i=17	j=1	0	0	0	0	0	0	0	0	0	0	0	14485	0	0	13229	0	0	0	0	0	0	14932	0	0	0	0	0	0	0	0
		j=5	14293	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	11959.2	0	0	0	12627	0	0	0	0	0	0	0	15038	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	i=18	j=1	0	12029	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13347.3	0	0	0	7158.5	0	0	0	0	12849	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	0	12945	12115	0	0	0	0	0	0	0	0	0	0	13451.2	0	0	0	0	0
		j=10	0	0	12761	0	15721	0	0	0	0	0	0	0	0	0	12945	0	0	0	7494.1	0	0	0	0	0	0	0	0	0	0	0
	i=19	j=1	0	0	0	0	0	0	12115	0	0	0	14519	11793.5	0	0	0	0	0	0	0	0	0	0	2441.4	13461	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	3574.5	0	14485	0	13558	13299	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	12714	0	0	0	0	0	0	0	0	0	0	0	0	0	13580	13037.6	0	0	0	0	4573.4	0
	i=20	j=1	0	0	0	0	0	0	0	0	0	15554	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	14445	0	0	0	0	0	0	0	0	0	0	15893	14619	13984	0	0	0	0	0	0	13986	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	14079	15789	0	0	0	0	0	0	0	0	14667	0	0	0	0	0	0	0	0	0
	i=21	j=1	0	0	15774	0	0	0	0	0	0	0	0	0	0	14203.1	14284	0	0	0	0	0	14792	0	0	0	0	13293	0	14875	0	0
		j=5	0	0	0	14483	0	0	0	0	0	0	15521	0	0	1700.9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13805	0
		j=10	0	0	0	0	0	0	15882	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13951	0	0	14505	0	0	0

Table 8
The Sensitivity analysis process on demand parameter.

Case	Demand	Objective function
Case1	$0.6 \times \delta^5_{qts}$	1454069
Case2	$0.7 \times \delta^5_{qts}$	1625317
Case3	$0.8 \times \delta^5_{qts}$	1896106
Case4	$0.9 \times \delta^5_{qts}$	2155737
Case5 (IP13)	δ^5_{qts}	2330884
Case6	$1.1 \times \delta^5_{qts}$	2517263
Case7	$1.2 \times \delta^5_{qts}$	2822604
Case8	$1.3 \times \delta^5_{qts}$	2994544
Case9	$1.4 \times \delta^5_{qts}$	3251304

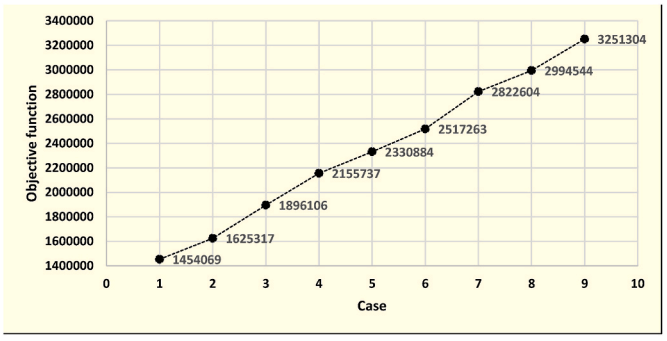


Fig. 4. The objective function trend for sensitivity analysis of demand parameter.

algorithm can arm managers and practitioners to make proper decisions in various contexts. It is noteworthy that the application scope of this model goes beyond pomegranates as it can cover other agricultural products for the sake of waste reduction, efficiency improvement, and a circular economy. It is of high importance to conduct more research and implement the model in other contexts so that its versatility and impact can be practically assessed.

8. Conclusion

By integrating the regeneration concept with SC network operations, positive environmental impacts can be achieved in addition to effective waste management and reduction of negative environmental impacts. The agricultural industry is one of the industries that embraces this concept. Applying regenerative principles in this industry leads to a reduction in the use of chemical fertilizers and environmental protection. Therefore, this paper, for the first time, presents a novel stochastic scenario-based MILP model for structuring a RSC network in the pomegranate industry, by employing the regeneration concept. The studied network was multi-product, multi-period, and multi-echelon, and included forward and reverse flows. In addition, this paper developed a Lagrangian relaxation algorithm to solve the problem in large scales. The data of a pomegranate industry in Iran was used to validate the proposed model and algorithm. Using probabilistic distribution functions to simulate the case study data, nine small-sized problems and five large-sized problems were generated. The results showed that the developed algorithm performs as well as GAMS in small-sized problems, except that it is faster. In addition, comparing the results of the proposed algorithm and GAMS on large-sized problems indicated the efficiency of the proposed algorithm.

For future research, we propose adding aspects like carbon emissions costs or fertilizer reduction advantages as the second objective function of the model. The model developed in this article is designed for an

agricultural SC in rural areas and quantifying these aspects is challenging. However, at national level or at multiple products with a robust data infrastructure, it is very interesting to consider these aspects in planning. Moreover, integrating IoT/blockchain features for traceability of organic fruit to consumers could also enhance the regenerative design of the chain and also provide high volume data for optimizing the chain according to real time data.

CRediT authorship contribution statement

Alireza Shamekhi Amiri: Writing – original draft, Supervision, Resources, Methodology, Conceptualization. **Parisa Karimi-Ashtiani:** Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Conceptualization. **Seyed Ali Modarresi:** Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Shahryar Sorooshian:** Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Conceptualization. **Hassan Mina:** Writing – original draft, Software, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge that the images presented in Fig. 2 were generated using artificial intelligence for enhanced visual appeal.

Appendix

Table A1
The selected suppliers for each time under scenario 2

Y^1_{its}		Time period (t)																												
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Supplier (i)	1					✓								✓			✓	✓				✓		✓		✓		✓	✓		✓
	2	✓							✓			✓	✓	✓	✓			✓	✓	✓	✓					✓			✓		
	3		✓			✓				✓									✓					✓				✓		✓	
	4	✓	✓	✓			✓	✓		✓	✓																✓				✓
	5			✓		✓	✓				✓		✓			✓				✓				✓	✓		✓			
	6					✓	✓		✓				✓		✓	✓	✓		✓						✓				✓	✓	✓
	7				✓			✓	✓							✓	✓		✓			✓									
	8	✓			✓	✓	✓		✓	✓				✓		✓								✓		✓	✓		✓	✓	
	9		✓		✓		✓		✓	✓	✓			✓	✓					✓	✓		✓						✓		
	10	✓	✓	✓					✓		✓		✓	✓	✓				✓	✓	✓		✓		✓	✓				✓	
	11	✓	✓					✓		✓				✓			✓	✓	✓		✓	✓		✓		✓		✓			
	12			✓			✓								✓						✓	✓	✓		✓						
	13	✓			✓			✓			✓					✓				✓	✓		✓	✓			✓		✓		
	14		✓	✓						✓								✓	✓	✓						✓		✓			✓
	15			✓				✓	✓		✓	✓		✓				✓			✓	✓				✓	✓	✓			
	16				✓		✓	✓				✓								✓								✓		✓	✓
	17	✓			✓				✓				✓			✓	✓						✓								
	18		✓	✓		✓						✓		✓	✓				✓	✓			✓				✓	✓			✓
	19									✓		✓	✓		✓		✓	✓					✓	✓	✓				✓	✓	✓
	20					✓					✓	✓	✓			✓	✓	✓				✓			✓						
	21		✓	✓	✓			✓				✓			✓	✓						✓			✓		✓	✓	✓	✓	

Table A2
The selected suppliers for each time under scenario 3

Y^1_{its}		Time period (t)																												
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Supplier (i)	1					✓								✓	✓		✓	✓				✓		✓		✓		✓	✓		✓
	2	✓		✓					✓			✓	✓	✓	✓			✓	✓	✓	✓					✓			✓		
	3		✓			✓				✓	✓								✓					✓		✓		✓		✓	
	4	✓	✓	✓	✓		✓	✓		✓	✓																✓				✓
	5			✓		✓	✓						✓			✓				✓				✓	✓		✓			
	6					✓	✓		✓			✓	✓		✓	✓	✓		✓				✓		✓				✓	✓	✓
	7				✓			✓	✓							✓	✓		✓			✓									
	8	✓			✓	✓	✓		✓	✓				✓		✓								✓		✓	✓		✓	✓	
	9		✓		✓		✓		✓	✓	✓		✓	✓				✓			✓		✓					✓	✓		
	10	✓	✓						✓		✓		✓	✓	✓				✓	✓	✓		✓		✓	✓				✓	
	11	✓	✓					✓		✓				✓			✓	✓	✓		✓	✓	✓	✓		✓		✓			
	12			✓			✓								✓						✓	✓	✓		✓			✓			
	13	✓			✓		✓	✓			✓					✓				✓	✓		✓	✓				✓		✓	✓
	14		✓	✓						✓							✓	✓	✓	✓						✓		✓			✓
	15			✓				✓	✓		✓	✓		✓				✓			✓	✓			✓	✓	✓	✓			
	16				✓		✓	✓		✓		✓								✓								✓		✓	✓

(continued on next page)

Table A2 (*continued*)

Y^1_{its}		Time period (t)																												
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
	17	✓			✓				✓				✓			✓	✓			✓	✓	✓	✓								
	18		✓	✓		✓						✓		✓	✓					✓			✓				✓	✓			✓
	19				✓	✓		✓		✓		✓	✓		✓		✓	✓	✓				✓	✓	✓				✓	✓	✓
	20					✓					✓	✓	✓	✓		✓	✓	✓				✓			✓		✓				
	21	✓	✓	✓	✓			✓				✓			✓	✓						✓			✓		✓		✓	✓	

Table A3
The amount of type 2 pomegranates purchased from suppliers by sorting centers in each time period under scenario 1

X^{1-2}_{qjits}			Time period (t)																																	
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
q=2	i=1	j=1	0	0	0	0	4661.6	0	0	0	0	0	0	0	0	0	0	8050	0	0	0	0	0	0	0	0	0	9301	8581	0	0
		j=5	0	0	0	4488.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9357.3	0	2208.8	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	8065	0	0	0	0	0	0	0	13.7	0	0	0	8571	0	0	0	0	9799
	i=2	j=1	9229	0	0	0	0	0	4309.1	0	0	0	0	5228.9	0	0	0	0	0	7370	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	7343	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8861	0	0	8555	0	0
		j=10	0	0	0	0	0	0	0	0	1027.3	7008	0	0	8912	0	0	9907	0	0	8638	0	0	0	0	0	0	0	0	0	0
	i=3	j=1	0	0	0	8873	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9702	0	0	0	0	0	0	0
		j=5	0	7913	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1403	0	644.7	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7033.9	0	0	0	0	0	0	0	0	0	0	0
	i=4	j=5	0	0	9478	0	0	0	8086	0	5602.9	8648	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7335	0	0	0	8348
		j=10	7825	9946	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	i=5	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7525	0	6153.6	0	0	0
		j=5	0	0	0	9788	0	0	0	0	0	0	9597	0	0	0	0	0	0	9301	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	8913	0	0	0	0	0	0	0	0	9738	0	0	0	0	0	0	0	0	0	0	0	0	0	8806	0
	i=6	j=1	0	0	0	0	0	0	0	0	0	0	0	0	9497	8200	7235	0	0	0	0	0	0	0	0	0	0	7624	0	0	0
		j=5	0	0	0	0	7850	0	9807	0	0	0	7303	0	0	1571	0	0	8839	0	0	0	0	0	0	4919.6	0	0	1568	7377	0
		j=10	0	0	0	0	5545.6	0	0	0	0	0	8563	0	0	0	0	0	0	0	0	0	0	0	0	4415.4	0	0	0	0	9863
	i=7	j=1	0	0	0	0	0	0	4832.9	0	0	0	0	0	0	0	6687.3	0	0	0	0	8543	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	4154.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	7619	0	0	8884	0	0	0	0	0	0	0	0	0	0	8935	0	0	0	0	0	0	0	0	0	0	0
	i=8	j=1	83.1	0	0	0	7669	0	0	9544	7609	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9976	0
		j=5	6984.9	0	0	8141	0	0	0	0	0	0	0	9139.5	0	6467.7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	7914	0	0	0	0	0	126.5	0	0	0	0	0	0	0	0	0	8737	0	0	9213	0	9084	0	0
	i=9	j=1	0	0	0	8084	0	6978	0	0	4522	8998	0	0	0	0	0	0	0	0	0	8277	0	0	0	0	0	4929.8	0	0	0
		j=5	0	0	0	0	1684	0	0	0	0	0	0	9020	0	0	0	0	0	0	0	0	0	4418	0	0	0	2111.2	0	0	0
		j=10	0	8034	0	0	0	0	9383	4739	0	0	0	0	0	0	0	0	0	0	544	0	4707	0	0	0	0	0	9306	0	0
	i=10	j=1	9760	7826	0	0	0	0	0	0	0	0	0	0	9547	0	0	0	8124	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	7608	0	9674	0	0	0	0	0	0	0	0	7044.4	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	5277.5	8211	0	0	0	0	0	0	0	0	7863	0	8553	8154	0	0	0	0	0
	i=11	j=1	7034.9	0	0	0	0	0	0	9445	0	0	0	9107	0	0	0	0	8556.7	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	8659	0	0	0	0	0	0	0	0	0	0	0	0	0	0	251.3	0	0	7426	0	7578	0	6421	0	9148	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8333	7184	0	0	8478	0	0	0	0	0	0	0	0	0	0
	i=12	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9783	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	9672	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9634	0	8610	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	8338	0	0	0	0	0	0	8678	0	0	0	0	0	0	0	0	0
	i=13	j=1	0	0	0	8838	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7221	0	0	0	0	0	0	0	0	8272	0
		j=5	0	0	0	0	0	9160	0	0	0	0	0	0	0	0	0	0	0	7864	0	0	8949	0	0	0	0	0	0	0	2392.5
		j=10	7948	0	7508	0	0	0	0	0	9870	0	0	0	0	8654	0	0	0	0	0	0	0	9736	0	0	0	0	0	0	0
	i=14	j=1	0	0	7594	0	0	0	0	0	0	0	0	0	0	0	0	0	0	349.7	0	0	0	0	0	7136	0	1233	0	0	9085
		j=5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9945	0	0	0	0	0	0	0	0	0	0	0	0
		j=6	0	8035	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	9823	0	0	0	0	0	0	0	9208	0	7682.3	0	0	0	0	0	0	0	6418	0	0	0
	i=15	j=1	0	0	0	0	0	6159.1	0	0	0	0	0	0	0	0	0	7403	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	9786	0	0	0	0	0	0	0	0	7281	9337	0	0	0	0	0	0	0	0	0
		j=10	0	0	8514	0	0	2126.9	0	0	9311	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9886	0	7377	0	0	0
	i=16	j=1	0	0	0	0	3557.9	9736	0	0	0	8217.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9947	0	0	0	0
		j=5	0	0	2514.4	0	0	0	0	0	0	3227.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7480	0	9729	0
		j=10	0	0	0	0	5813.1	0	0	0	0	0	0	0	0	0	0	0	0	9457	0	0	0	0	0	0	0	0	0	0	0
	i=17	j=1	0	0	0	0	0	0	0	0	0	0	8168	0	0	9881	8643	0	0	0	0	0	10000	0	0	0	0	0	0	0	0
		j=10	0	0	0	9733	0	0	0	8041	0	0	0	0	0	0	1067	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	i=18	j=1	0	8981.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7975	0	0	0	8852	0	0	0	0	9855	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	7738	9820	0	0	0	0	0	0	0	0	0	0	9005	0	0	0	0	0
		j=10	0	0	9647	0	7040	0	0	0	0	0	0	0	0	0	0	0	0	9879	0	0	0	0	0	0	0	0	0	0	0
	i=19	j=1	0	0	0	0	0	0	9004	0	0	0	7366	0	0	0	0	0	0	0	0	0	0	0	0	3094.9	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	8446	0	9825	0	8059	9504	0	0	0	0	0	0	0	6042.1	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	9351	0	0	0	0	0	0	0	0	0	0	0	0	6080.8	9016	0	0	0	0	0	8344	0
	i=20	j=1	0	0	0	0	0	0	0	0	9136	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	9343	0	0	0	0	0	0	0	0	0	8289	9995	8656	0	0	0	0	0	0	4689.1	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	3.7	9710	0	0	0	0	0	0	0	0	8466	0	0	0	0	0	0	0	0	0
	i=21	j=1	0	0	9132	0	0	0	0	0	0	0	0	0	1681.4	7539	0	0	0	0	0	6233.1	0	0	0	0	8231	0	7399	0	0
		j=5	0	0	0	8952	0	0	0	0	0	8274	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7848	0
		j=10	0	0	0	0	0	0	1077.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9994	0	0	0	0	0

Table A4
The amount of type 3 pomegranates purchased from suppliers by sorting centers in each time period under scenario 1

X^{1-2}_{qijts}			Time period (t)																													
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
q=3	i=1	j=1	0	0	0	0	3911.4	0	0	0	0	0	0	0	0	0	0	0	4423	0	0	0	0	0	0	0	0	0	4470	3715	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	0	3800	0	0	0	0	0	0	0	0	0	4463	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4538	0	0	0	3964	0	0	0	0	4906
	i=2	j=1	3751	0	0	0	0	0	0	4802	0	0	0	0	4548	0	0	0	0	0	4911	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4042	0	0	897.5	0	0
		j=10	0	0	0	0	0	0	0	0	0	3607	4701	0	0	0	0	0	0	0	0	3646	0	0	0	0	0	0	0	0	0	0
	i=3	j=1	0	0	0	0	4602	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3902	0	0	0	0	0	0	0
		j=5	0	1475.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3665	0	4308	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4060	0	0	0	0	0	0	0	0	0	0	0	0
	i=4	j=5	0	0	4674	0	0	0	3862	0	4316	4329	0	0	0	0	0	0	0	0	0	0	0	0	0	3743	0	0	0	0	4868
		j=10	3801	4373	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	i=5	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4899	0	4297	0	0	0	0	0
		j=5	0	0	0	0	4003	0	0	0	0	0	0	4335	0	0	0	0	0	0	4916	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	4976	0	0	0	0	0	0	0	0	4847	0	0	0	0	0	0	0	0	0	0	0	0	3568	0	0
	i=6	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	4908	1833	3843	0	0	0	0	0	0	0	0	0	0	3544	0	0	0
		j=5	0	0	0	0	0	3746	0	3797	0	0	0	4604	0	0	0	0	0	143.6	0	0	0	0	4223	0	0	0	0	2895.7	0	0
		j=10	0	0	0	0	4970	0	0	0	0	0	4805	0	0	0	3016	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4978
	i=7	j=1	0	0	0	0	0	0	4592	0	0	0	0	0	0	0	0	3908	0	0	0	0	4710	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	3315.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	4399	0	0	0	0	0	0	0	0	0	0	0	0	4561	0	0	0	0	0	0	0	0	0	0	0	0	0
	i=8	j=1	4743	0	0	0	4527	0	0	4665	4060	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4906	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	0	4697	0	4588	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	519.9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4976	0	0	3695	0	4183	0	0
	i=9	j=1	0	0	0	4599	0	0	0	0	3679	4521	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4572	0	0	0
		j=5	0	0	0	0	0	4714	0	0	0	0	0	0	4143	0	0	0	0	0	0	0	0	2618	0	0	0	0	0	0	0	0
		j=10	0	4828	0	0	0	0	0	4901	0	0	0	0	0	4788	0	0	0	0	0	4970	0	0	0	0	0	0	4777	0	0	0
	i=10	j=1	3849	3532	0	0	0	0	0	0	0	0	0	0	0	2379.2	0	0	0	4684	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	3914	0	4336	0	0	0	0	0	0	0	0	3750	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	4572	3718	0	0	0	0	0	0	0	0	3568	0	3671	4506.7	0	0	0	0	0
	i=11	j=1	4492	0	0	0	0	0	0	0	4808	0	0	0	3962	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	4053	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4735	0	0	0	0	4226	0	3721	0	4853	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3982	4912	0	0	4647	0	0	0	0	0	0	0	0	0	0
	i=12	j=1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1496.5	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	4980	0	0	4785	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3944	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4930	0	0	0	0	0	0	0	0	0
	i=13	j=1	0	0	0	3881	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3550	0	0	0	0	0	0	0	4567	0	0
		j=5	0	0	0	0	0	0	3528	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4321	0	0	0	0	0	0	4222	0
		j=10	629.4	0	3091.9	0	0	0	0	0	0	3510	0	0	0	0	4102	0	0	0	0	0	0	0	1397.6	0	0	0	0	0	0	0
	i=14	j=1	0	0	4890	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3829	0	0	0	0	2820.3
		j=5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3675	0	0	0	0	0	0	0	0	0	0	0	0
		j=6	0	4507	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	4046	0	0	0	0	0	0	0	1361.9	0	3453.6	0	0	0	0	0	0	0	0	0	0	0
	i=15	j=1	0	0	0	0	0	0	4513	0	0	0	0	0	0	0	0	0	4391	0	0	0	3639	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	3963	0	0	0	0	0	0	0	0	4931	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	4577	0	0	0	0	0	0	4511	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4799	4810	0	0	0	0
	i=16	j=1	0	0	0	0	0	0	4981	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	111.7	0	0	0	0
		j=5	0	0	0	4591	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4397	0	3895
		j=10	0	0	0	0	0	4545	0	0	0	0	0	0	0	0	0	0	0	0	4269	0	0	0	0	0	0	0	0	0	0	0
	i=17	j=1	0	0	0	0	0	0	0	0	0	0	0	3972	0	0	3925	4575	0	0	0	0	0	3680	0	0	0	0	0	0	0	0
		j=5	4832	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=10	0	0	0	1105.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	i=18	j=1	0	4950	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4783	0	0	0	4221	0	0	0	0	3169.2	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	0	2918.6	3736	0	0	0	0	0	0	0	0	0	0	117.3	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4457	0	0	4333	0	0	0	0	0	3412.7	0	0	0	0	0
	i=19	j=1	0	0	0	0	0	0	0	0	0	0	4637	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		j=5	0	0	0	0	0	0	0	0	0	0	0	4763	0	4638	0	0	4407	0	0	0	0	0	0	1370.1	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	3567.2	0	0	0	0	0	0	0	0	0	0	0	0	3598	4514	0	0	0	0	0	0	0
	i=20	j=5	0	0	0	0	4220	0	0	0	0	0	0	0	0	0	441.2	2384.8	3893	0	0	0	0	0	4899	0	0	0	0	0	0	0
		j=10	0	0	0	0	0	0	0	0	0	3176.7	4834	3656	0	0	0	0	0	0	0	0	3243.3	0	0	0	0	0	0	0	0	0
	i=21	j=1	0	0	4542	0	0	0	0	0	0	0	0	0	4742	3968	0	0	0	0	0	0	3722	0	0	0	3632	0	3813	0	0	0
		j=5	0	0	0	4756	0	0	0	0	0	0	3829	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4651	0	0
		j=10	0	0	0	0	0	0	4349	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3861	0	0	3577	0	0	0

Table A5
The amount of pomegranates shipped from sorting centers to pomegranate market in each time period under scenario 1

X^{2-3}_{qjts}		Time period (t)																														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
q=1	j=1	16904.4	23422.7	19763.1	31791.3	24286.7	24647.5	10622.9	29456.4	24389.4	26634.4	24520.1	22303.6	27022.3	25550.8	23351.4	15812	15745	24389.7	35532.1	27610.8	18340.1	24552.7	14500.2	24056.7	20858.4	25875.3	21497.8	23920.1	14454.7	19170.8
	j=5	25029.4	19525.4	37008.5	19204	19589.8	26886.6	24464.1	12198	22224.5	22998.9	27656	24586.7	18061.2	25620	35120.1	23915	30788.9	15817.1	23477.9	23101.1	31077.2	23945.5	38107.9	23731.4	33706.3	23245.5	25832.9	28169.3	29715.8	24852.4
	j=6	0	12777.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2018.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	j=10	34146.2	28224	22998.4	31894.7	36393.5	23556	37342.9	32345.5	24536.1	21576.7	25903.8	31589.7	34216.6	32589.1	25528.4	36153	23647.7	30423.2	18130	25048.1	25942.6	29381.9	18721.9	25411.8	17445.3	33359.2	36549.4	28080.6	33709.5	26216.8
q=2	j=1	22273.7	14339.6	14270.1	14437.3	18090.2	8988.9	21243.1	15942.3	18408	15471.4	13295.6	6968.7	12230.9	17682.3	21858.2	19252	13184	21035.5	6586.2	21569	12606.5	16083.9	8277.4	9060.5	6088.2	20759	21601.2	20138.2	15568.6	7751
	j=5	5959.3	8025.2	14199.9	16728.5	13022.1	23515.3	14713.8	18402.1	4780.2	18107.5	19476.5	21624.4	22094.9	16760.5	13930.3	13528.2	17368.5	16240.4	11342.6	15524.1	22285.1	19623.7	8349.8	20698.7	13038.1	13940.8	8739.3	17082	13539.5	17463.9
	j=6	0	6855.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	j=10	13457	15340	21900	14804.2	10737.7	19315.9	10313.1	14865.6	20401.8	17241.1	13287.9	12786.9	13994.1	14717.2	15691.5	8019.8	22437.5	13624.2	23051.2	15067	14638.4	15912.3	23452.8	19590.8	22703.7	7860.2	11769.5	15689.8	14631.9	16775
q=3	j=1	14849.8	7481.8	8319.8	7480	11502.7	0	12425	8350.7	11067.5	3987.9	4090.2	3503.6	7506.5	10610.7	8579.1	10872.5	7774.7	8350.7	4331.9	4451.4	10647.6	6969.3	3441.9	4321.3	3377.5	7092.5	10771.2	9766.4	8355.9	2487.7
	j=5	4262.2	4876.6	8515.6	8244.8	7253.3	11683.2	6248.9	9996.3	3807.1	7643.2	5470.8	13488.6	13723.9	7305.9	4516.8	2103.6	7321.3	7544.9	7644.1	4349.5	0	9599.7	7664.4	9254.9	6847.6	3405.1	7513.6	4670.2	10456.8	11453.8
	j=6	0	3975.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	j=10	3908	8116	6764.6	4855.2	4383.9	8856.8	3836.2	4323.1	6715.5	13058.9	12649	7257.8	3279.6	4223.4	10554.1	7443.9	5534.1	7604.4	10634	11699	11212.4	6321	9603.7	6643.8	11704.9	10512.4	3155.2	7903.4	3147.3	8718.5

Table A6
The amount of pomegranates shipped from sorting centers to juice extraction centers in each time period under scenario 1

X^{2-4}_{qjkts}			Time period (t)																												
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
q=1	j=1	k=2	0	4174.3	0	5665.7	0	0	1893.2	372.2	0	4746.6	0	3974.8	1378.2	632.5	0	2817.9	0	4346.6	6332.4	0	0	0	0	0	3717.3	0	3831.2	0	0	0
		k=8	3012.6	0	3522.1	0	4328.3	4392.5	0	4877.4	4346.6	0	4369.9	0	3437.6	3921	4161.6	0	2806	0	0	4920.7	3268.5	4375.7	2584.2	4287.3	0	4611.4	0	4262.9	2576	3416.5
	j=5	k=2	0	0	0	882.6	0	0	0	0	3937.3	0	5529.5	2319	0	0	6258.9	0	5487.1	3.6	1289.1	3784.4	5538.4	0	0	0	0	0	0	1861.5	0	4429.1
		k=8	4460.6	3479.7	6595.5	2539.9	3701.9	4580.9	4359.9	2197.3	0	4296.7	0	1264.1	3218.8	4565.9	0	4262	0	2815.2	3227.6	0	0	4364.1	6694.7	4232.7	6003.6	4142.7	4603.8	3158.7	5295.8	0
	j=6	k=2	0	2277.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	359.7	0	0	0	0	0	0	0	0	0	0	0	0	0
		k=8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	j=10	k=2	5570.5	0	4098.7	0	6485.9	3698.3	0	5764.5	0	0	0	0	6097.9	5807.9	0	610.5	0	0	0	0	0	5236.3	1559.7	4528.8	0	5945.1	0	5004.4	6001.9	0
		k=8	514.9	5030	0	5684.1	0	499.7	6655.1	0	4372.7	3845.3	4616.5	5629.8	0	0	4549.6	5832.6	4214.4	5421.9	3231	4463.9	4623.4	0	1776.8	0	3109	0	6513.6	0	5.7	4672.2
q=2	j=1	k=2	0	2467.9	0	2484.7	1717	0	3655.9	0	2373.7	2662.6	397.1	1199.3	0	0	1182.4	3313.3	0	3620.2	1133.5	0	0	0	0	54.4	1047.8	727.3	3717.6	0	0	0
		k=8	3833.3	0	2455.9	0	1396.4	1547	0	2743.7	794.3	0	1891.1	0	2104.9	3043.1	2579.4	0	2269	0	0	3712	2169.6	2768.1	1424.6	1504.9	0	2845.4	0	3465.8	2679.4	1334
	j=5	k=2	0	0	0	0	0	0	1304.3	0	822.7	0	2699.8	0	0	0	2397.4	0	2666.4	0	0	1069.1	3835.3	414.8	0	0	1426.6	0	0	0	0	1411.3
		k=8	1025.6	2433.3	1391.7	2879	3468.1	2820	1227.9	3167	0	3768.4	0	3721.6	3802.5	2884.5	0	2650.9	0	2795	3554.7	0	0	2962.5	1437	3562.2	817.3	2399.2	1859.2	2584.7	2330.2	1594.2
	j=6	k=2	0	1179.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		k=8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	j=10	k=2	2316	0	3769	0	1848	3324.3	0	2558.4	0	2243.1	0	0	2408.4	2532.8	0	0	0	0	0	2446.7	101	2738.5	4036.2	3371.6	0	1352.8	1026.4	2700.2	2518.1	0
		k=8	0	2640	0	2547.8	0	0	1774.9	0	3511.2	724.1	2286.8	2200.6	0	0	2700.5	1380.2	3861.5	2344.7	3967.1	146.3	2418.2	0	0	0	3907.3	0	999.1	0	0	2887
q=3	j=1	k=2	0	918.1	813.7	1000	0	0	1661	0	0	533.1	0	468.4	0	1418.5	1146.9	1453.5	0	1116.3	579.1	0	0	0	0	0	451.5	0	1439.9	0	0	0
		k=8	1985.2	82.1	298.5	0	1537.7	0	0	1116.3	1479.5	0	546.8	0	1003.5	0	0	0	1039.3	0	0	595.1	1423.4	931.7	460.1	577.7	0	948.2	0	1305.6	1117.1	332.6
	j=5	k=2	0	0	0	0	0	0	0	0	508.9	0	918.8	0	0	0	593	0	506.8	0	1021.9	581.5	0	1283.3	0	0	915.4	0	0	0	0	1531.2
		k=8	569.8	651.9	1138.4	1102.2	969.7	1561.8	871.4	1300.3	0	1021.8	0	1615.7	1834.7	987.5	0	281.2	472	1008.6	0	0	0	0	1024.6	1237.2	0	455.2	1004.4	624.3	1397.9	0
	j=6	k=2	0	531.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		k=8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	j=10	k=2	522.4	0	904.3	0	586.1	1184	0	577.9	0	0	0	0	438.4	564.6	0	0	0	0	0	0	0	845	1283.9	888.2	0	1405.3	0	1056.6	420.7	0
		k=8	0	1085	0	649.1	0	0	512.8	0	897.8	1745.8	1691	970.2	0	0	1410.9	995.1	739.8	1016.6	1421.6	1564	1498.9	0	0	0	1564.8	0	421.8	0	0	1165.5

Table A7
The amount of pomegranate juice shipped from juice extraction centers to market in each time period under scenario 1

X^{4-5}_{kts}	Time period (t)																														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
k=2	4624.9	6351.8	5272.1	5518.1	5850.3	4513.6	4683	5100.1	4203.5	5602	5249.9	4378.8	5677.6	6026	6368.2	4507.3	4961	4997.7	5695.8	4334.9	5211.1	5784.8	3783.9	4863.6	4157.2	5186.8	5508.3	5842.5	4917.4	4054.4
k=8	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1	8471.1

Table A8
The amount of pomegranate waste shipped from juice extraction centers to composting centers in each time period under scenario 1

X^{4-6}_{klts}		Time period (t)																													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
k=2	l=2	3784	4341.1	4313.6	4514.8	4786.6	3693	0	4172.8	0	4341.1	4295.4	0	0	4341.1	0	3687.8	4059	4089.1	0	3546.8	0	0	3095.9	3979.3	0	0	4506.8	4341.1	4023.3	3317.2
	l=5	0	0	0	0	0	0	3831.5	0	3439.2	0	0	3582.7	4341.1	0	4341.1	0	0	0	4660.2	0	4263.6	4733.1	0	0	3401.3	4243.7	0	0	0	0
	l=8	0	855.8	0	0	0	0	0	0	0	242.4	0	0	304.2	589.2	869.3	0	0	0	0	0	0	0	0	0	0	0	0	439.1	0	0
k=8	l=2	1109.9	1109.9	1109.9	936.2	664.4	1109.9	4941.4	1109.9	4549.1	1109.9	1109.9	4692.6	5451	1109.9	5451	1109.9	1109.9	1109.9	5451	1109.9	5373.5	5451	1109.9	1109.9	4511.2	5353.6	944.2	1109.9	1109.9	1109.9
	l=5	5821	5821	5821	5821	5821	5821	1989.5	5821	2381.8	5821	5821	2238.3	1479.9	5821	1479.9	5821	5821	5821	1160.8	5821	1557.4	1087.9	5821	5821	2419.7	1577.3	5821	5821	5821	5821
	l=8	0	0	0	173.7	445.5	0	0	0	0	0	0	0	0	0	0	0	0	0	319.1	0	0	392	0	0	0	0	165.7	0	0	0

Table A9
The amount of compost shipped from composting centers to farmers in each time period under scenario 1

X^{6-1}_{lts}	Time period (t)																														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
l=2	1957.6	2180.4	2169.4	2180.4	2180.4	1921.1	1976.6	2113.1	1819.6	2180.4	2162.1	1877	2180.4	2180.4	2180.4	1919.1	2067.6	2079.6	2180.4	1862.7	2149.4	2180.4	1682.3	2035.7	1804.5	2141.4	2180.4	2180.4	2053.3	1770.8
l=5	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4	2328.4
l=8	0	342.3	0	69.5	178.2	0	0	0	0	96.9	0	0	121.7	235.7	347.7	0	0	0	127.6	0	0	156.8	0	0	0	0	66.3	175.6	0	0

Table A10
The amount of pomegranates stored at warehouses of sorting centers in each time period under scenario 1

X_{qjts}		Time period (t)										
		2	5	7	8	10	11	14	16	19	22	24	27
q = 1	j = 5	0	1182.4	0	131.6	1110.4	4481.6	0	0	1866	542.4	18.9	0
q = 2		6113.5	7129.3	0	0	3789.1	0	0	1874.9	9312.1	0	0	2063.7
q = 3		0	0	269.7	0	0	1402.4	80.6	0	0	0	0	0

Data availability

Data will be made available on request.

References

Abad, A.R.K.K., Barzinpour, F., Pishvaee, M.S., 2023. Toward circular economy for pomegranate fruit supply chain under dynamic uncertainty: a case study. Comput. Chem. Eng. 178, 108362.

Aguilar, G., Paulino, S., 2025. Different approaches for transformation of agri-food system in times of climate change: agroecology and regenerative agriculture.

Agroecology and Sustainable Food Systems. https://doi.org/10.1080/21683565.2025.2469066.
Alfina, K.N., Ratnayake, R.C., Wibisono, D., Basri, M.H., Mulyono, N.B., 2024. Prioritizing performance indicators for the circular economy transition in healthcare supply chains. Circ. Econ. Sustain. https://doi.org/10.1007/s43615-024-00436-9.
Allaoui, H., Guo, Y., Choudhary, A., Bloemhof, J., 2018. Sustainable agro-food supply chain design using two-stage hybrid multi-objective decision-making approach. Comput. Oper. Res. 89, 369–384.
Amir, S., Salehi, N., Roci, M., Sweet, S., Rashid, A., 2024. Toward a circular economy: a guiding framework for circular supply chain implementation. In: Sustainable Supply Chains: a Research-based Textbook on Operations and Strategy. Springer International Publishing, Cham, pp. 379–404.
Bag, S., Rahman, M.S., 2024. Navigating circular economy: unleashing the potential of political and supply chain analytics skills among top supply chain executives for environmental orientation, regenerative supply chain practices, and supply chain viability. Bus. Strat. Environ. 33 (2), 504–528.
Banasik, A., Kanellopoulos, A., Claassen, G.D.H., Bloemhof-Ruwaard, J.M., van der Vorst, J.G., 2017. Closing loops in agricultural supply chains using multi-objective optimization: a case study of an industrial mushroom supply chain. Int. J. Prod. Econ. 183, 409–420.
Batista, L., Gong, Y., Pereira, S., Jia, F., Bittar, A., 2019. Circular supply chains in emerging economies–a comparative study of packaging recovery ecosystems in China and Brazil. Int. J. Prod. Res. 57 (23), 7248–7268.
Blanchard, C., Harris, P., Pocock, C., McCabe, B.K., 2023. Food and garden organic waste management in Australia: Co-benefits for regional communities and local government. Sustainability 15 (13), 9901.
Bless, A., Davila, F., Plant, R., 2023. A genealogy of sustainable agriculture narratives: implications for the transformative potential of regenerative agriculture. Agric. Hum. Val. 40 (4), 1379–1397.
Bolívar, J., Cantillo, V., Miranda, P., 2025. Agri–food supply chain design for perishable products: application to small-scale farmers. Oper. Res. 25 (2), 26.
Brown, M.M., Haselsteiner, E., Apró, D., Kopeva, D., Luca, E., Pulkkinen, K.L., Rizvanolli, B., 2018. Restorative to regenerative: an exploration in progressing a paradigm shift in built environment thinking, from sustainability to restorative sustainability and on to regenerative sustainability. https://hdl.handle.net/10863/18060.
Caron, P., Ferrero y de Loma-Osorio, G., Nabarro, D., Hainzelin, E., Guillou, M., Andersen, I., et al., 2018. Food systems for sustainable development: proposals for a profound four-part transformation. Agron. Sustain. Dev. 38, 1–12.
Cheraghalipour, A., Roghanian, E., 2022. A bi-level model for a closed-loop agricultural supply chain considering biogas and compost. Environ. Dev. Sustain. 38 (41). https://doi.org/10.1007/s13593-018-0519-1.
Cheraghalipour, A., Paydar, M.M., Hajiaghaei-Keshteli, M., 2018. A bi-objective optimization for citrus closed-loop supply chain using Pareto-based algorithms. Appl. Soft Comput. 69, 33–59.
Chouhan, V.K., Khan, S.H., Hajiaghaei-Keshteli, M., 2021. Metaheuristic approaches to design and address multi-echelon sugarcane closed-loop supply chain network. Soft Comput. 25 (16), 11377–11404.
Day, C., Cramer, S., 2022. Transforming to a regenerative US agriculture: the role of policy, process, and education. Sustain. Sci. 17 (2), 585–601.
Dipu, M.A., Jones, N.A., Aziz, A.A., 2022. Drivers and barriers to uptake of regenerative agriculture in southeast Queensland: a mental model study. Agroecology and Sustainable Food Systems 46 (10), 1502–1526.
Duncan, J., Carolan, M., Wiskerke, J.S., 2020. Regenerating food systems: a social-ecological approach. In: Routledge Handbook of Sustainable and Regenerative Food Systems. Routledge, pp. 1–11.
Fasihi, M., Tavakkoli-Moghaddam, R., Najafi, S.E., Hajiaghaei-Keshteli, M., 2021. Developing a bi-objective mathematical model to design the fish closed-loop supply chain. Int. J. Eng. 34 (5), 1257–1268.
Fisher, M.L., 2004. The Lagrangian relaxation method for solving integer programming problems. Manag. Sci. 50 (12_Suppl. ment), 1861–1871.
Genovese, A., Acquaye, A.A., Figueroa, A., Koh, S.L., 2017. Sustainable supply chain management and the transition towards a circular economy: evidence and some applications. Omega 66, 344–357.
Gholian-Jouybari, F., Hajiaghaei-Keshteli, M., Bavar, A., Bavar, A., Mosallanezhad, B., 2023b. A design of a circular closed-loop agri-food supply chain network—A case study of the soybean industry. Journal of Industrial Information Integration 36, 100530.
Gholian-Jouybari, F., Hajiaghaei-Keshteli, M., Smith, N.R., Calvo, E.Z.R., Mejía-Argueta, C., Mosallanezhad, B., 2024. An in-depth metaheuristic approach to design a sustainable closed-loop agri-food supply chain network. Appl. Soft Comput. 150, 111017.
Gholian-Jouybari, F., Hashemi-Amiri, O., Mosallanezhad, B., Hajiaghaei-Keshteli, M., 2023a. Metaheuristic algorithms for a sustainable agri-food supply chain considering marketing practices under uncertainty. Expert Syst. Appl. 213, 118880.
Gholipour, A., Sadegheih, A., Mostafaeipour, A., Fakhrzad, M.B., 2024. Designing an optimal multi-objective model for a sustainable closed-loop supply chain: a case study of pomegranate in Iran. Environ. Dev. Sustain. 26 (2), 3993–4027.
Gholipour, A., Sadeghieh, A., Mostafaeipour, A., Fakhrzad, M.B., 2023. Designing a closed-loop supply chain network with a combined algorithm solution method: a case study of pomegranates. Journal of Industrial and Systems Engineering 14 (4), 234–269.
Gibbons, L.V., 2020. Regenerative—The new sustainable? Sustainability 12 (13), 5483.
Gilani, H., Sahebi, H., 2021. Optimal design and operation of the green pistachio supply network: a robust possibilistic programming model. J. Clean. Prod. 282, 125212.
Goodarzian, F., Ghasemi, P., Gonzalez, E.D.S., Tirkolaee, E.B., 2023. A sustainable-circular citrus closed-loop supply chain configuration: Pareto-based algorithms. J. Environ. Manag. 328, 116892.
Gordon, E., Davila, F., Riedy, C., 2022. Transforming landscapes and mindscapes through regenerative agriculture. Agric. Hum. Val. 39 (2), 809–826.
Gordon, E., Davila, F., Riedy, C., 2023. Regenerative agriculture: a potentially transformative storyline shared by nine discourses. Sustain. Sci. 18 (4), 1833–1849.
Gosnell, H., 2022. Regenerating soil, regenerating soul: an integral approach to understanding agricultural transformation. Sustain. Sci. 17 (2), 603–620.
Govindan, K., Asgari, F., Fard, F.S.N., Mina, H., 2024. Application of IoT technology for enhancing the consumer willingness to return E-waste for achieving circular economy: a Lagrangian relaxation approach. J. Clean. Prod. 459, 142421.
Govindan, K., Nosrati-Abarghooee, S., Nasiri, M.M., Jolai, F., 2022. Green reverse logistics network design for medical waste management: a circular economy transition through case approach. J. Environ. Manag. 322, 115888.
Heidari-Fathian, H., Pasandideh, S.H.R., 2018. Green-blood supply chain network design: robust optimization, bounded objective function & lagrangian relaxation. Comput. Ind. Eng. 122, 95–105.
Jayasinghe, S.L., Thomas, D.T., Anderson, J.P., Chen, C., Macdonald, B.C., 2023. Global application of regenerative agriculture: a review of definitions and assessment approaches. Sustainability 15 (22), 15941.
Khan, M., Ajmal, M.M., Jabeen, F., Talwar, S., Dhir, A., 2023. Green supply chain management in manufacturing firms: a resource-based viewpoint. Bus. Strat. Environ. 32 (4), 1603–1618.
Liao, Y., Kaviyani-Charati, M., Hajiaghaei-Keshteli, M., Diabat, A., 2020. Designing a closed-loop supply chain network for citrus fruits crates considering environmental and economic issues. J. Manuf. Syst. 55, 199–220.
Loza Adaui, C.R., 2025. The principle of regeneration in circular economy: revitalising for resilience. In: Circular Economy in Sustainable Supply Chains: a Global Perspective on Challenges, Concepts and Cases. Springer Nature Switzerland, Cham, pp. 49–60.
Mehrabi, S., Mina, H., Sorooshian, S., 2025. Optimization of a resilient circular closed-loop supply chain network under uncertainty. Clean. Eng. Technol., 100995
Mirzaei, M.G., Gholami, S., Rahmani, D., Goodarzian, F., Khanchehzarrin, S., 2024. Robust optimization in an agricultural closed-loop supply chain network design with a price and freshness-dependent demand: hybrid rat with particle swarm optimization algorithm. Environ. Dev. Sustain. https://doi.org/10.1007/s10668-024-05296-9.
Mohan, S.V., Hemalatha, M., Amulya, K., Velvizhi, G., Chiranjeevi, P., Sarkar, O., et al., 2020. Decentralized urban farming through keyhole garden: a case study with circular economy and regenerative perspective. Materials Circular Economy 2, 1–12.
Morseletto, P., 2020. Restorative and regenerative: exploring the concepts in the circular economy. J. Ind. Ecol. 24 (4), 763–773.
Mosallanezhad, B., Hajiaghaei-Keshteli, M., Triki, C., 2021. Shrimp closed-loop supply chain network design. Soft Comput. 25, 7399–7422.
Mpanga, I.K., Schuch, U.K., Schalau, J., 2021. Adaptation of resilient regenerative agricultural practices by small-scale growers towards sustainable food production in north-central Arizona. Current Research in Environmental Sustainability 3, 100067.
Nwaogu, C., Agidi, V.A., Bridget, D.E., Okorondu, J.N., Olushola, B., Edokpa, D., et al., 2024. Unveiling the prospects of regenerative agriculture in promoting Soil-C stocks, food security, and climate change mollification and resilience. In: International Conference on Geo-science and Geospatial Technologies: Ecosystem Restoration and Natural Resources Management. Springer Nature Switzerland, Cham, pp. 117–123.
Park, J., Conca, K., Finger, M., 2008. The Crisis of Global Environmental Governance: towards a New Political Economy of Sustainability. Routledge.
Pascucci, S., 2020. Circular food economies. In: Routledge Handbook of Sustainable and Regenerative Food Systems. Routledge, pp. 318–335.
Pascucci, S., Duncan, J., 2017. From pirate islands to communities of hope: reflections on the circular economy of food systems. In: Sustainable Food Futures. Routledge, pp. 186–200.
Rahemi, H., Torabi, S.A., Avami, A., Jolai, F., 2020. Bioethanol supply chain network design considering land characteristics. Renew. Sustain. Energy Rev. 119, 109517.
Rajabi-Kafshgar, A., Gholian-Jouybari, F., Seyedi, I., Hajiaghaei-Keshteli, M., 2023. Utilizing hybrid metaheuristic approach to design an agricultural closed-loop supply chain network. Expert Syst. Appl. 217, 119504.
Reyhani Yamchi, H., Jabarzadeh, Y., Govindan, K., Amoozad Mahdiraji, H., 2024. A triple bottom line approach for designing a sustainable closed-loop supply chain network in fruit industry: a metaheuristic solution approach. J. Oper. Res. Soc. 75 (10), 1925–1948.
Roghanian, E., Cheraghalipour, A., 2019. Addressing a set of meta-heuristics to solve a multi-objective model for closed-loop citrus supply chain considering CO2 emissions. J. Clean. Prod. 239, 118081.
Röös, E., Wood, A., Säll, S., Hatab, A.A., Ahlgren, S., Hallström, E., et al., 2023. Diagnostic, regenerative or fossil-free-exploring stakeholder perceptions of Swedish food system sustainability. Ecol. Econ. 203, 107623.
Salehi-Amiri, A., Zahedi, A., Akbapour, N., Hajiaghaei-Keshteli, M., 2021. Designing a sustainable closed-loop supply chain network for walnut industry. Renew. Sustain. Energy Rev. 141, 110821.
Salehi-Amiri, A., Zahedi, A., Gholian-Jouybari, F., Calvo, E.Z.R., Hajiaghaei-Keshteli, M., 2022. Designing a closed-loop supply chain network considering social factors; a case study on avocado industry. Appl. Math. Model. 101, 600–631.
Schattman, R.E., Rowland, D.L., Kelemen, S.C., 2023. Sustainable and regenerative agriculture: tools to address food insecurity and climate change. J. Soil Water Conserv. 78 (2), 33A–38A.

Schreefel, L., De Boer, I.J.M., Timler, C.J., Groot, J.C.J., Zwetsloot, M.J., Creamer, R.E., et al., 2022. How to make regenerative practices work on the farm: a modelling framework. Agric. Syst. 198, 103371.
Schreefel, L., Schulte, R.P., De Boer, I.J.M., Schrijver, A.P., Van Zanten, H.H.E., 2020. Regenerative agriculture–the soil is the base. Global Food Secur. 26, 100404.
Schulte, L.A., Dale, B.E., Bozzetto, S., Liebman, M., Souza, G.M., Haddad, N., et al., 2022. Meeting global challenges with regenerative agriculture producing food and energy. Nat. Sustain. 5 (5), 384–388.
Sehnem, S., Jabbour, C.J.C., Pereira, S.C.F., de Sousa Jabbour, A.B.L., 2019. Improving sustainable supply chains performance through operational excellence: circular economy approach. Resour. Conserv. Recycl. 149, 236–248.
Seydanlou, P., Jolai, F., Tavakkoli-Moghaddam, R., Fathollahi-Fard, A.M., 2022. A multi-objective optimization framework for a sustainable closed-loop supply chain network in the olive industry: hybrid meta-heuristic algorithms. Expert Syst. Appl. 203, 117566.
Taherifar, M., Hasani, N., Zokaee, M., Aghsami, A., Jolai, F., 2024. A scenario-based sustainable dual-channel closed-loop supply chain design with pickup and delivery considering social conditions in a natural disaster under uncertainty: a real-life case study. Environ. Dev. Sustain. 26 (8), 19443–19490.
Tavana, M., Kian, H., Nasr, A.K., Govindan, K., Mina, H., 2022. A comprehensive framework for sustainable closed-loop supply chain network design. J. Clean. Prod. 332, 129777.
Wilson, K.R., Hendrickson, M.K., Myers, R.L., 2024. A buzzword, a "win-win", or a signal towards the future of agriculture? A critical analysis of regenerative agriculture. Agric. Hum. Val. 42, 257–269.
Yavari, M., Zaker, H., 2019. An integrated two-layer network model for designing a resilient green-closed loop supply chain of perishable products under disruption. J. Clean. Prod. 230, 198–218.
Yuvaraj, M., Jothi Basu, R., Prabhu, B.V., Boppana, S.B., Ganesh Kumar, C., 2025. Design of multi-echelon multi-fruit supply chain network considering multi-compartment reefer trucks to ensure food security. Br. Food J. 127 (5), 1756–1787.
Zeunert, J., 2018. Challenges in agricultural sustainability and resilience: towards regenerative practice. In: Routledge Handbook of Landscape and Food. Routledge, pp. 231–252.